

Multi-crosslinked nanocomposite hydrogels for hydrocarbon recovery: Design, properties, and performance

Xuechen Tang^{a,b,*}, Xinji Du^c, Yiqiang Li^{b,c}, Tao Song^c, Jinxin Cao^c, Zheyu Liu^{b,c}, Shilin Zeng^c

^a Exploration and Development Research Institute of PetroChina Southwest Oil & Gas Field Company, Chengdu, Sichuan, 610041, PR China

^b State Key Laboratory of Petroleum Resources and Engineering, China University of Petroleum (Beijing), Beijing, 102249, PR China

^c College of Petroleum Engineering, China University of Petroleum (Beijing), Beijing, 102249, PR China

ARTICLE INFO

Keywords:

Multi-crosslinked
Dual-nanocomposite hydrogel
Conformance control
Hydrocarbon recovery
Gelling property

ABSTRACT

Water channeling and reservoir heterogeneity remain critical bottlenecks in enhanced oil recovery (EOR), especially in low-grade reservoirs where water flooding exhibit poor efficiency. Hydrogels have emerged as promising conformance control materials; however, traditional hydrogels suffer from inherent brittleness, inadequate energy dissipation, and insufficient thermal stability, restricting their applicability in harsh reservoir environments. To address these limitations, this study developed a dual-nanocomposite (DNC) hydrogel by synthesizing surface-functionalized nano-silica and incorporating graphene oxide. These nanomaterials were integrated into a multi-crosslinked polymer network via *in-situ* free radical polymerization, functioning as reinforcing fillers and multifunctional crosslinkers to enhance the gel structure. Comprehensive characterizations and tests were conducted to evaluate the hydrogel's performance: Rheological analyses demonstrated a prolonged gelation time (16.3% extension of the initial gelation time) and superior viscoelasticity (a storage modulus of 34,360 Pa), along with excellent thermal stability (retaining structural integrity for over 180 days at elevated temperatures). One-dimensional core experiments revealed a remarkable plugging rate of 95.59%, while visualization tests and heterogeneous core flooding tests confirmed a 40.29% improvement in sweep efficiency and a 9.56-percentage-point increase in ultimate oil recovery. These results validate that the DNC hydrogel effectively diverts fluid flow from high-permeability to low-permeability zones, achieving robust conformance control. This work establishes DNC hydrogels as advanced functional materials, offering sustainable and efficient solutions for the development of unconventional hydrocarbon resources.

Nomenclature

Symbols	Definition
A	Cross-sectional area
c	Concentration of the NaOH solution
d	Average particle size
f	Surface functionalization degree
F_{τ}	Residual resistance factor
G'	Storage modulus
G''	Loss modulus
H	Total height of the sample
h	Selected height of the sample
k	Permeability
k_0	Permeability before gel treatment

(continued on next column)

(continued)

k'	Permeability after gel treatment
m	Mass of SiO ₂
N_A	Avogadro constant
ΔP	Injection pressure difference
$\Delta P_{\max}$	Breakthrough pressure difference
Q	Water production rate
Q_s	Optical parameter given by Mie theory
r_i	Internal radius of the measurement cell
S	Specific surface area of SiO ₂
T	Transmission flux
t	Test time
ΔT	Delta transmitted intensity
$\Delta T_{\max}$	Maximum delta transmitted intensity

(continued on next page)

* Corresponding author. Exploration and Development Research Institute of PetroChina Southwest Oil & Gas Field Company, Chengdu, 610041, Sichuan, PR China.

E-mail address: tangxuechen@petrochina.com.cn (X. Tang).

<https://doi.org/10.1016/j.geoen.2026.214591>

Received 10 February 2025; Received in revised form 28 April 2026; Accepted 25 May 2026

Available online 25 May 2026

2949-8910/© 2026 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

(continued)

T_0	Transmittance of the continuous phase
t_0	Initial gelation time
t_c	Gelation completion time
t_{dec}	Decorrelation time
V	Volume of the NaOH solution consumed to raise the pH from 4.0 to 9.0
V_0	Volume of the NaOH solution consumed for pristine SiO ₂ particles
V_m	Volume of the NaOH solution consumed for functionalized SiO ₂ particles
v_{sed}	Sedimentation velocity
Greek symbols	
δ_{peak}	Thickness of the subsidence peak
ζ	Zeta potential
η	Plugging rate
λ	Photon mean free path
μ	Fluid viscosity
σ	Surface density
ϕ	Volume concentration of dispersion phase
Abbreviations	
CNT	Carbon nanotube
DLS	Dynamic Light Scattering
DNC	Dual-nanocomposite
EI	Elasticity index
EOR	Enhanced oil recovery
GO	Graphene oxide
GQDs	Graphene quantum dots
GSC	Gel Strength Code
LOWESS	Locally weighted scatterplot smoothing
MLS	Multiple Light Scattering
MSD	Mean Square Displacement
MVI	Macroscopic viscosity index
MS-DWS	Multi-Speckle Diffusing Wave Spectroscopy
NC	Nanocomposite
OOIP	Original oil in place
PAM	Polyacrylamide
PV	Porosity volume
SAOS	Small amplitude oscillatory shear
SP	Surfactant-polymer
SLB	Solid-liquid balance
TDS	Total dissolved solids
TMS	Tetramethylsilane
TOCNF	Tempo-treated carboxylated nanocellulose
TSI	Turbiscan Stability Index

1. Introduction

Exponential population growth and rapid industrialization have established crude oil as the primary energy source for humanity, which will continue to dominate the global energy market for the foreseeable future (Pal and Mandal, 2020; Tang et al., 2025). Consequently, as the exploration and development of oil and gas resources continue to deepen, low-grade reservoirs become the key breakthrough areas in boosting both reserves and efficiency in the oil industry (Lele et al., 2018; Xu et al., 2024; Xiao et al., 2025). Although water flooding is one of the most cost-effective and widely used methods for conventional reservoirs, its efficiency is significantly reduced in low-grade reservoirs. Long-term water flooding failure is primarily caused by the stratigraphic anisotropy and complexity in these reservoirs, such as fractures and high-permeability zones, as well as the unique properties of crude oil, including high viscosity and solid-phase deposition caused by high asphaltene and wax content (Seright and Brattekas, 2021; Wang et al., 2021; Tang et al., 2024). These factors aggravate the formation of high-conductivity channels, which drastically reduce sweep efficiency. It is estimated that 65%–77% of the original oil in place (OOIP) remains in the unswept area of reservoirs (Zhou et al., 2022; Tang et al., 2023; Zhang et al., 2024). Therefore, strengthening research into enhanced oil recovery (EOR) technologies, including chemical water shutoff, conformance control, and permanent plugging of high-permeability zones, is essential for extending reservoir lifespans and ensuring stable oil production (Bai et al., 2015).

Hydrogels are three-dimensional, crosslinked polymeric networks with excellent water absorption capacity, extensively applied in oilfields

for conformance control due to their reliable performance and cost-effectiveness (Bai et al., 2015; Salunkhe et al., 2021; Du et al., 2025; Yuan et al., 2025). In particular, *in-situ* gels are formed directly within the reservoir by using polymers (or monomers) and crosslinkers. The gelant solutions are injected into targeted areas, where they undergo crosslinking underground, resulting in gel formation that effectively seals the reservoir. The low viscosity of the gelant solutions enables them to travel further from the injection well through high-permeability pathways, achieving deep conformance control (Qin et al., 2023). Despite extensive research and application, these gels still exhibit several limitations when used. On the one hand, traditional hydrogels with a single, non-reversible covalent crosslinked network are typically brittle and exhibit limited responsiveness to external forces, as they lack mechanisms for effective energy dissipation (Du et al., 2019a; Fukao et al., 2020; Lu et al., 2021). On the other hand, they often suffer from poor thermal stability due to their sparse and uneven network structures, further restricting their application under harsh reservoir conditions (Almeida et al., 2024). Adverse reservoir environments frequently lead to the degradation or dehydration of these conventional gels, ultimately leading to the failure of in-depth conformance control treatments.

To tackle the numerous challenges associated, nanomaterials have emerged as critical research focuses, serving as reinforcing agents in the development of nanocomposite hydrogels (Qin et al., 2023; Karchoubi et al., 2024; Lu et al., 2024). These hydrogels combine hydrophilic polymer networks, crosslinked via covalent bonds or physical interactions, with nanoparticles to form nanoscale complex structures. The integration of nanomaterials into hydrogels consistently improves their structural integrity, mechanical properties (e.g., tensile strength, fracture energy, elasticity modulus), thermal properties (e.g., conductivity and heat capacity), and functional behaviors such as swelling, release rates, and rheological characteristics (Martín et al., 2019; Karchoubi et al., 2024). For instance, Zhang et al. (2021) developed an acidic, shear-tolerant nanosheet-enhanced cross-linked hydrogel system by synthesizing an organic zirconium crosslinker and incorporating surface-functionalized MoS₂ nanosheets, significantly enhancing the gel's rheological and shear-resistant properties. Similarly, Liu et al. (2023) synthesized a highly stretchable and self-healing hydrogel for sustainable CO₂ shutoff by introducing a synergistic mechanism of tempo-treated carboxylated nanocellulose (TOCNF) and Fe³⁺. Different types (metallic, inorganic, carbon-based, polymeric, or hybrid) and dimensionalities (0D, 1D, and 2D) of nanomaterials bring distinct enhancement mechanisms to composites (Kumar et al., 2022; Natalya et al., 2022; Karchoubi et al., 2024; Shen et al., 2024). For example, Tang et al. (2022) compared the interfacial properties and rheological behavior of four polymeric nanofluids (nano-SiO₂, nano-CaO, GO, and CNT), and found that each exhibited unique advantages: nano-SiO₂ excelled in reducing interfacial tension, Carbon nanotube (CNT) showed superior wettability alteration and rheological properties, GO provided balanced performance across metrics, and nano-CaO demonstrated moderate improvements. Martín et al. (2019) noted that the impact of graphene quantum dots (GQDs) on hydrogel structure was found to be neutral, as they do not constitute part of the polymeric network, contrasting with graphene and GO, which significantly alter the structure and properties of hydrogels through different mechanisms. Therefore, compared to single nanomaterials, their combination can exert synergistic effects across diverse types and dimensions within the polymer matrix, collectively boosting the overall performance of hydrogels (Zhang et al., 2013; Du et al., 2016; Xu et al., 2019; Dems et al., 2021). Du et al. (2019b) synthesized a PAM/SiO₂/TiO₂ hybrid hydrogel, where nanoparticles served as hybrid crosslinkers, enhancing toughness, elasticity, and structural stability through covalent and non-covalent interactions, resulting in high mechanical strength (330.50 kPa tensile, 6.22 MPa compressive) and excellent UV-shielding and anti-fogging properties. Xu et al. (2019) demonstrated that introducing Laponite and SiO₂ nanoparticles to form a dual nano-composite structure

significantly contributes to the enhancement of mechanical properties and malleability for hydrogels. Although promising advancements have been made, studies on these multi-nanomaterial-reinforced hydrogels for hydrocarbon recovery in oilfields remain limited.

Herein, inspired by previous research, a dual-nanocomposite (DNC) hydrogel for hydrocarbon recovery was synthesized. Acrylamide and acrylic acid served as comonomers to construct the hydrogel backbone via *in-situ* free radical polymerization, while surface-functionalized SiO₂ and graphene oxide were incorporated as multifunctional crosslinkers and reinforcing fillers. The integration of these nanomaterials significantly enhanced the gel network through improved crosslink density, hydrogen bonding, and nanoparticle-polymer interactions. The resulting DNC hydrogel demonstrated excellent gelling properties, including a controllable gelation time and a highly resilient crosslinked network, making it adaptable to complex reservoir environments. It maintained structural integrity for over 180 days at elevated temperatures, with a storage modulus exceeding 34 kPa, outperforming both traditional hydrogels and single-nanoparticle systems. These advancements render the DNC hydrogel highly injectable for smooth reservoir deployment, targetable for selective conformance control, blockable with a remarkable plugging rate of 95.59%, and durable for long-term applications in challenging environments. The hydrogel's superior plugging capability translates into enhanced conformance control and improved EOR efficiency. In heterogeneous cores, the ultimate oil recovery increased by 9.56 percentage points compared to cases without gel treatment. These findings underscore the transformative potential of DNC hydrogels in addressing critical challenges in EOR, paving the way for more efficient and sustainable exploitation of low-grade reservoirs.

2. Experimental

2.1. Materials

Sodium chloride (NaCl) and hydrochloric acid (HCl) were purchased from Sinopharm Chemical Reagent Co., Ltd (Shanghai, China). Acrylic acid (AA), ammonium hydroxide (NH₃·H₂O), and sodium hydroxide (NaOH) were supplied by Shanghai Aladdin Biochemical Technology Co., Ltd. Acrylamide (AM), anhydrous ethanol (C₂H₅OH), ammonium persulfate (APS), and polyacrylamide (PAM, 1.0 × 10⁷ g mol⁻¹) were obtained from Shanghai Macklin Biochemical Technology Co., Ltd. Other chemicals, including 3-(methacryloyloxy) propyl trimethoxysilane (C₁₀H₂₀O₅Si, MPS), 4-styrenesulfonic acid sodium salt (C₈H₇NaO₃S, SSS), 3-(2-(methacryloyloxy) ethyl) dimethylammonio-propane-1-sulfonate (C₁₁H₂₁NO₅S, DMAPS), N,N'-Methylenebisacrylamide (C₇H₁₀N₂O₂, MBA), and deuterated dimethyl sulfoxide (DMSO-d₆, C₂D₆SO) were procured from Tianjin Heowns Biochemical Technology Co., Ltd. Silicon dioxide (SiO₂) was sourced from Nanjing XFANO Materials Tech. Co., Ltd., while graphene and its derivative graphene oxide (GO) were acquired from Suzhou Tanfeng Tech. Co., Ltd. Aviation kerosene was supplied by Jinan Xinke Chemical Co., Ltd. Industrial surfactants and hydrophobically associating polymers were provided by PetroChina Xinjiang Oilfield Branch. All reagents were used as received without further purification, and distilled water was used throughout.

The dehydrated waxy crude oil from Shen 84-An 12 block, Shenyang Oilfield, was adopted in Sec. 2.6, with a density of 0.865 g·cm⁻³ (20 °C, 0.1 MPa), wax content of 39.13%, and viscosity ranging from several mPa·s to hundreds of thousands of mPa·s between room temperature and formation temperature (70 °C) (Tang et al., 2025). Synthetic brine from Shenyang Oilfield with a total dissolved solids (TDS) content of 3384.0 mg·L⁻¹ was employed for solution preparation (Table 1).

Table 1
Shenyang Oilfield synthesized water.

Composition	Na ⁺ +K ⁺	Mg ²⁺	Ca ²⁺	Cl ⁻	SO ₄ ²⁻	HCO ₃ ⁻	CO ₃ ²⁻	TDS
Concentration/mg·L ⁻¹	960.5	9.1	20.1	238.2	24.0	2051.4	106.3	3384.0

2.2. Polymerizable modification and characterization of nano-SiO₂

2.2.1. Surface functionalization of SiO₂ nanoparticles

SiO₂ nanoparticles were functionalized with MPS in a 250 mL three-neck flask equipped with mechanical stirring, reflux condenser, and nitrogen protection. 1 g SiO₂ was ultrasonically dispersed in a water/anhydrous ethanol mixture (volume ratio 1:10), then mixed with MPS and several drops of NH₃·H₂O. The reaction was carried out at 50 °C with 500 rpm stirring for 24 h. The product was centrifuged, filtered, washed with anhydrous ethanol, and dried at 60 °C for 48 h to obtain surface-functionalized silica (MPS-SiO₂).

The surface functionalization degree (*f*) of SiO₂ nanoparticles was regulated by modifier dosage and determined via the following procedure. 0.4 g SiO₂ was mixed with 10 mL anhydrous ethanol and 100 mL 20 wt% NaCl solution, sonicated for 10 min, and adjusted to pH 4.0 using 0.1 mol·L⁻¹ HCl or NaOH. NaOH solution was added dropwise to reach pH 9.0 and held for 20 s. Silanol surface density (*σ*) and functionalization degree (*f*) were calculated by Equations (1) and (2) (Xu et al., 2014):

$$\sigma = \frac{cVN_A \times 10^{-3}}{mS} \quad (1)$$

$$f = \frac{V_0 - V_m}{V_0} \times 100\% \quad (2)$$

Here, *c* represents the NaOH concentration, mol·L⁻¹; *V* denotes the volume of the NaOH consumed from pH 4.0 to 9.0, mL; *N_A* is the Avogadro constant, 6.02 × 10²³; *m* is SiO₂ mass, g; *S* is specific surface area, nm²·g⁻¹; *V₀* and *V_m* are NaOH consumptions for pristine and modified SiO₂, mL, respectively.

2.2.2. Characterization and dispersion stability test

FTIR spectra of pristine and functionalized SiO₂ were recorded on a Thermo Fisher Scientific Nicolet iS20 spectrometer (USA). TEM observations were performed using a JEOL JEM-2100F microscope. Contact angles were measured with an SL200KS analyzer. For phase distribution analysis, pristine and modified SiO₂ were dispersed in a 1:1 (v/v) kerosene/water mixture, shaken vigorously, and left standing for 3 days to observe the particle distribution. Colloidal stability of 0.1 wt% SiO₂ suspensions was evaluated using a Turbiscan LAB Expert (Formulation, France) at 25 °C for 6 h with 2-min scanning intervals. Based on Multiple Light Scattering (MLS), Turbiscan enables non-destructive determination of dispersion stability, eliminating subjective interference and identifying instability mechanisms (e.g., flocculation, aggregation, migration) (Ren et al., 2018). The instrument yields two signals: transmitted light (for clear liquids) and backscattered light (for turbid/high-concentration samples). Transmitted light was adopted in this study, and its flux *T* follows the Lambert-Beer law as a function of photon mean free path *λ* (Shen et al., 2022):

$$T(\lambda, r_i) = T_0 e^{-\frac{2r_i}{\lambda}} \quad (3)$$

$$\lambda(d, \phi) = \frac{2d}{3\phi Q_s} \quad (4)$$

where, *r_i* is the cell radius, mm; *T₀* denotes the transmittance of the continuous phase; *d* is the average particle size, μm; *φ* refers to the volume concentration of the dispersion phase; and *Q_s* is the Mie optical parameter.

The Turbiscan Stability Index (*TSI*) quantifies sample stability by

integrating transmission/backscattering variations over specified heights, directly reflecting the overall dispersion stability. Calculated via Eq. (5), *TSI* compares each height-specific scan to its predecessor and normalizes the resultant value by the total sample height to eliminate dependence on sample volume in the measurement tube (Xiong et al., 2020):

$$TSI = \sum_i \frac{\sum_h |\text{Scan}_i(h) - \text{Scan}_{i-1}(h)|}{H} \quad (5)$$

where, h and H are the selected height and the total sample height, respectively. A higher *TSI* value denotes a more unstable system.

2.3. Morphology and characterization of graphene oxide

GO nanosheets were prepared via a modified Hummers' method. Morphology was characterized by SEM (ZEISS Sigma 300, Germany) and TEM. Chemical groups and phase structure were analyzed by FTIR and XRD (Scintag XDS2000, Cu K α radiation, 1.5405 Å, 40 kV, 20 mA). Zeta potential (ζ) was measured at 25 °C using a Malvern nanosizer.

2.4. Preparation and characterization of dual-nanocomposite hydrogel

2.4.1. Gel preparation

DNC hydrogels were synthesized via *in-situ* free radical polymerization (Fig. 1). Monomers (AM, AA, SSS, DMAPS) were dissolved in distilled water, followed by MBA crosslinker. MPS-SiO₂ and GO dispersion were added sequentially, and the mixture was sonicated for 30 min. APS initiator was added dropwise, and the total mass was adjusted to 40 g with deionized water. The solution was degassed in an ice bath for 3 min under nitrogen, then transferred into a hydrothermal reactor and cured at 60 °C for more than 12 h. Conventional polyacrylamide gel (base hydrogel) and silica-reinforced hydrogel (NC hydrogel) were prepared using the same procedure without GO/SiO₂ or without GO, respectively. The detailed formulation is listed in Table 2.

2.4.2. Characterization

The gel samples were characterized by SEM for microstructure observation and EDS for elemental composition analysis. FTIR was employed to assess chemical structural differences among the various gel samples. For investigating the interaction between GO and polymer backbone, ¹H NMR spectra of PAM and PAM/GO were recorded in DMSO-*d*₆ on a Bruker Ascend TM 600 MHz spectrometer (Germany) using tetramethylsilane (TMS) as the internal standard.

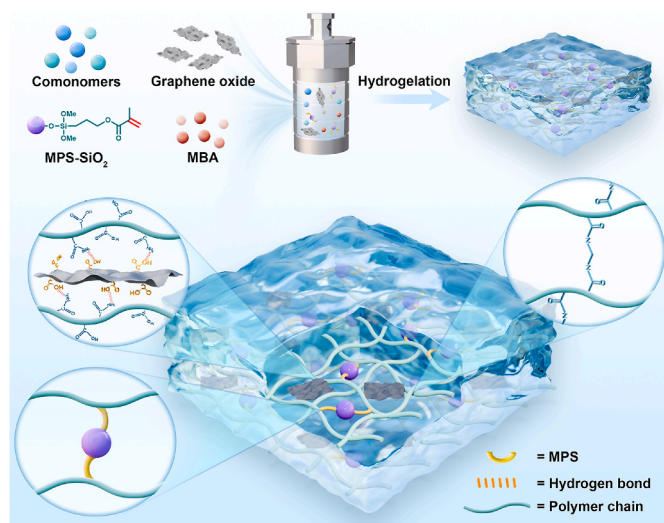


Fig. 1. Schematic diagram of the preparation process of DNC hydrogel.

2.5. Gelling properties and thermal stability of dual-nanocomposite hydrogel

The DG26.7 double-gap measuring system (Physica MCR301, Anton Paar, Austria) was utilized to compare the initial viscosity of various gelant solutions against distilled water at 6 r·min⁻¹ and ambient room temperature. Subsequently, the solutions were transferred to the CC27 concentric cylinder system for small amplitude oscillatory shear (SAOS) tests to investigate the gelation process at 60 °C, with strain fixed at 0.1% and frequency at 0.1 Hz to ensure linear viscoelasticity.

In this study, microrheology was utilized to investigate the viscoelastic behavior of soft materials at the micron scale by tracking the displacement of dispersed probe particles (Tisserand et al., 2012; Zhang et al., 2016; Shen et al., 2021). This method provides insight into the gelling properties and thermal stability of nanocomposite hydrogels. The Rheolaser optical microrheometer (Formulation Rheolaser Master) was employed to measure the Mean Square Displacement (*MSD*) of probe particles across a spatial range of 0.1–100 nm and a temporal scale spanning from 10⁻² to 10⁵ s, grounded in the principle of multi-speckle diffusing wave spectroscopy (MS-DWS) (Tisserand et al., 2012). Driven by thermal energy, particle displacement is ~10¹² times weaker than the macroscopic mechanical stress applied by conventional rheometers, enabling *in-situ*, non-invasive monitoring of material structural evolution over a broad time scale. For gelling property evaluation, *MSD* curves were continuously recorded as a function of decorrelation time (t_{dec}) at 60 °C for 12 h. For long-term stability assessment, gel strength was graded A to J (weak to strong) per the Sydansk Gel Strength Code (GSC), with microrheological tests conducted every 5 days over 180 days (Sydansk and Argabright, 1988).

2.6. Conformance control and EOR capacities of dual-nanocomposite hydrogel

This section aims to investigate the conformance control and EOR potential of DNC hydrogels in complex reservoir environments. Fig. 2a depicts the experimental setup, with all experiments conducted at a constant temperature of 60 °C. Artificial cores were prepared to match the reservoir properties of the Shen 84-An 12 block based on the artificial core fabrication technology of full pore space distribution.

2.6.1. Plugging performance estimation

Plugging performance was evaluated using one-dimensional sandstone cores (2.5 cm diameter, 10 cm length). The experimental procedure was as follows: (1) evacuate the core and saturate it with synthesized water; (2) calculate the core porosity volume (PV) based on saturated water content; (3) sequentially inject 1 PV synthesized water, 0.15 PV gelant solution (with 24 h of gelation time), and 2 PV synthesized water at 0.3 mL·min⁻¹, recording water production and injection pressure difference; (4) calculate core permeability k , plugging rate η , and residual resistance factor F_{rr} using Eqs. (6)–(8) (Dai et al., 2023).

$$k = \frac{Q\mu L}{A\Delta P} \quad (6)$$

$$\eta = \frac{k_0 - k'}{k_0} \times 100\% \quad (7)$$

$$F_{rr} = \frac{k_0}{k'} \quad (8)$$

where, Q is the water production rate, mL·s⁻¹; μ is the fluid viscosity, mP·s; A is the core cross-sectional area, cm²; ΔP is the injection pressure difference, 10⁻¹ MPa; k_0 and k' are the permeability before and after gel treatment, μm^2 , respectively.

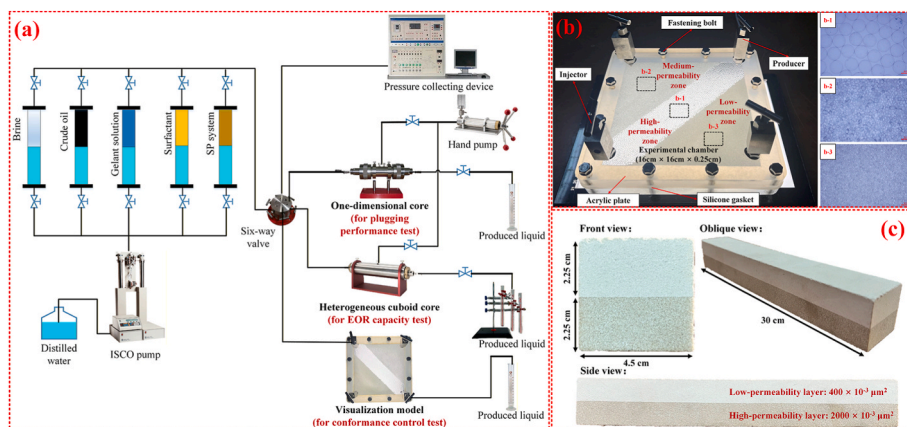
2.6.2. Conformance control ability observation

A custom visualization model (inner cavity: 16 cm × 16 cm ×

Table 2

Recipe for preparation of DNC hydrogel.

Materials	Water phase							Additives	
	AM	AA	SSS	DMAPS	MBA	APS ^a	Deionized water	SiO ₂	GO ^b
Mass/g	1	1	0.145	0.196	0.087	0.1	37.232	0.2	0.04

^a APS was configured as 15 wt% aqueous solution during use.^b GO was prepared as 5 mg/mL aqueous dispersion before use.**Fig. 2.** Schematic diagram for conformance control and EOR capacities. (a) Experimental apparatuses; and images for (b) the visualized model and (c) the heterogeneous cuboid core.

0.25 cm) was used to evaluate the conformance control capability of the DNC hydrogel (Fig. 2b). Glass beads (diameters: 2.5, 0.6, 0.4 mm) simulated high-, medium-, and low-permeability zones in heterogeneous reservoirs, with the high-permeability zone aligned along the injector–producer line and the other two zones distributed bilaterally. After evacuation and saturation with synthetic brine, the model was saturated with crude oil to establish irreducible water saturation. Water flooding was performed until no oil was produced, followed by chemical flooding until ultimate oil recovery is achieved. The displacement rate was maintained at $0.1 \text{ mL} \cdot \text{min}^{-1}$ throughout the experiment, with the process recorded in real-time using a high-speed camera. Three slug combinations were tested: (1) 0.5 PV surfactant slug, (2) 0.5 PV SP system slug, and (3) combined slug: 0.15 PV gelant solution (24 h gelation time post-injection) + 0.5 PV SP system slug.

2.6.3. EOR potential evaluation

A double-layer cuboid core with a permeability ratio of 5 and an average permeability of $1250 \times 10^{-3} \mu\text{m}^2$ (Fig. 2c) was fabricated to assess the EOR performance of heterogeneous reservoirs with interlayer channeling under different displacement strategies. For initial condition setup, the core was evacuated, saturated with synthetic water, and then flooded with dehydrated oil at $0.1 \text{ mL} \cdot \text{min}^{-1}$. Once no more brine was produced (marking full oil saturation), the core was aged for 48 h. Two parallel experimental groups were then subjected to the following procedures: (1) water flooding until the water cut of produced fluid reached 95%; (2) injection of 0.15 PV gelant solution into one group, followed by a 24-h gelation period, with the other group kept untreated as a control; (3) injection of 0.5 PV SP system; (4) post-flooding with water to achieve the ultimate oil recovery. Throughout the experiments, the injection rate was maintained at $0.5 \text{ mL} \cdot \text{min}^{-1}$. The dynamic variation of pressure difference and liquid production data were continuously recorded to clarify the effect of the DNC hydrogel slug on the EOR potential of heterogeneous reservoirs.

3. Results and discussion

3.1. Polymerizable modification of nano-SiO₂ and its performance

3.1.1. Characterization

MPS was employed to treat the particle surface to reinforce interfacial linkage between the polymer matrix and inorganic fillers, optimizing dispersibility, and strengthen the bonding of nano-SiO₂ to the matrix. As depicted in Fig. 3a, the infrared spectrum of SiO₂ displays characteristic peaks at 3440 cm^{-1} and 944 cm^{-1} , indicating the stretching and bending vibrations of the Si–OH group, respectively. The bending vibration of the H–O–H bond is observed at 1633 cm^{-1} , while peaks at 1101 cm^{-1} and 800 cm^{-1} represent the asymmetric and symmetric stretching vibrations of the Si–O–Si bond, respectively (Doan et al., 2024). Additionally, the bending vibration of Si–O–Si is identified at approximately 471 cm^{-1} (Deng et al., 2025). The introduction of silane results in notable changes in the infrared spectra. For instance, new absorption peaks at 2962 cm^{-1} and 2894 cm^{-1} are attributed to the asymmetric and symmetric stretching vibrations of methylene ($-\text{CH}_3$), respectively, while the bending vibration of $-\text{CH}_3$ is identified at 1456 cm^{-1} . Peaks at 1716 cm^{-1} and 1298 cm^{-1} are associated with the C=O and C–O–C bonds, confirming the presence of ester groups. Moreover, the intensity of the absorption peak at 3440 cm^{-1} , which corresponds to the Si–OH bond, is reduced, indicating a decrease in hydroxyl groups on the SiO₂ surface. These observations confirm the successful grafting of γ -MPS onto the SiO₂ surface, providing reactive groups (C=C) for subsequent composite reactions. As shown in Fig. 3b, unmodified SiO₂ exhibited strong hydrophilicity with complete dispersion in the aqueous phase, while MPS-modified SiO₂ was predominantly distributed at the oil–water interface, showing amphiphilicity. Fig. 3c further demonstrated that MPS-SiO₂ had reduced agglomeration and significantly improved dispersibility, which was attributed to the replacement of hydrophilic silanol groups with lipophilic MPS moieties, thus weakening interparticle hydrogen bonding.

3.1.2. Regulation of polymerizable modification degree

To maximize the bonding strength between SiO₂ particles and the

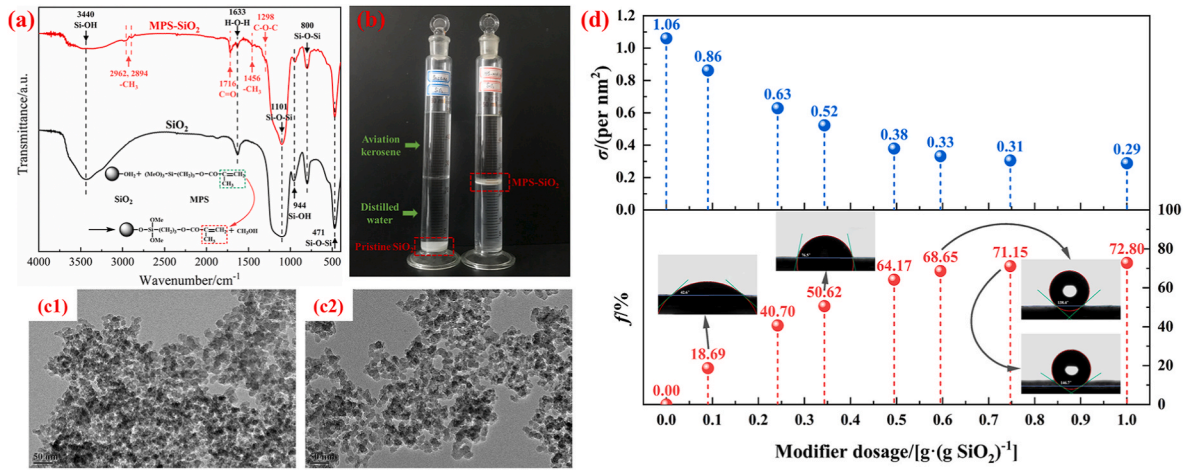


Fig. 3. Characterization of MPS-SiO₂. (a) FTIR spectra, (b) distributions in oil-water phases, (c) TEM micromorphologies, and (d) relationship between surface functionalization degree and modifier dosage.

polymer matrix, the particle surface functionalization degree f needs to be optimized. As illustrated in Fig. 3d, modifier dosage exhibits a clear

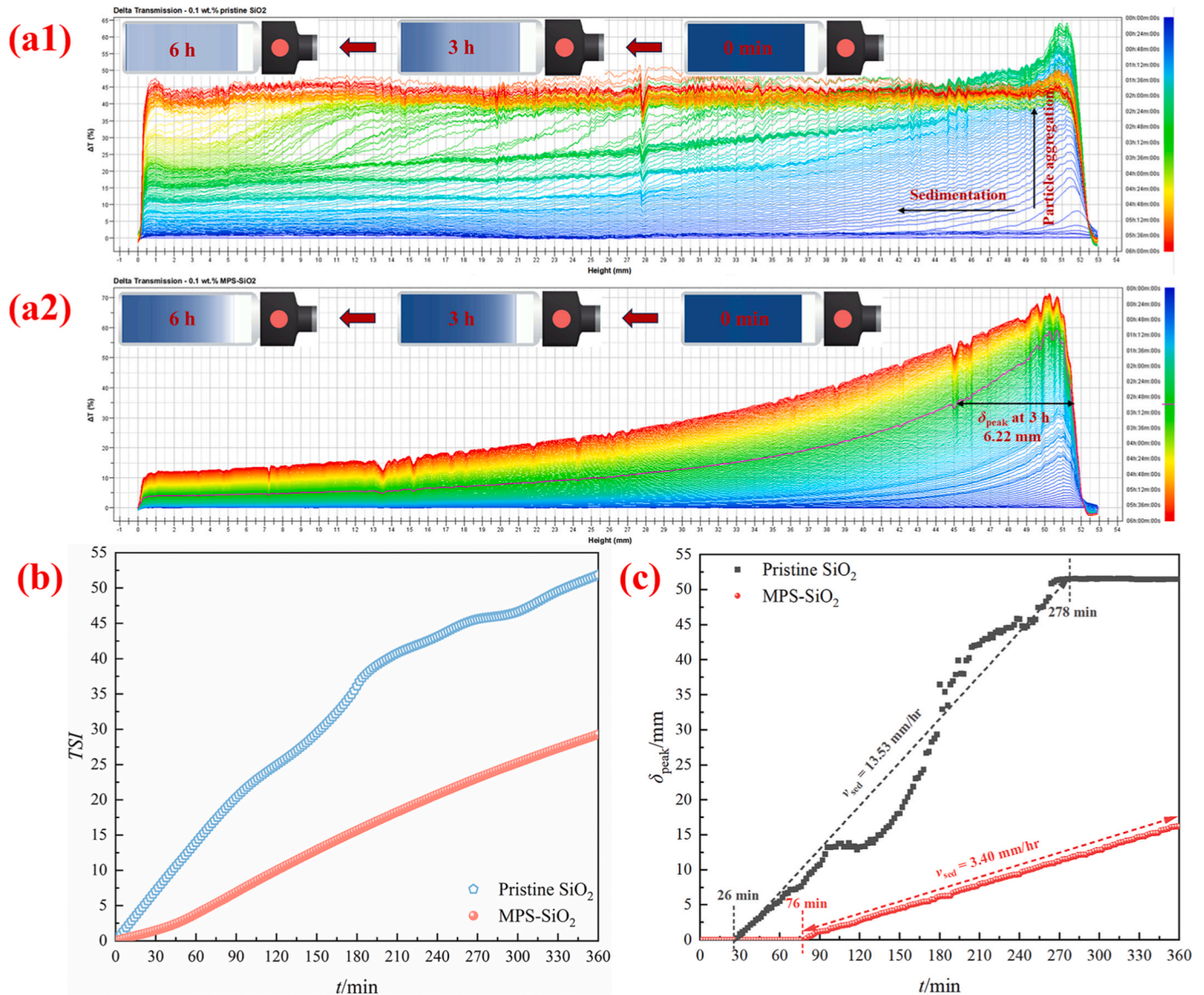


Fig. 4. Dispersion stability of 0.1 wt% SiO₂ dispersion systems. (a) Transmission spectra, (b) TSI values, and (c) migration rate.

correlation with both σ and f . With increasing MPS concentration, the silanol group content on the SiO₂ surface decreases, and f increases accordingly. When the modifier dosage reaches 0.5954 g·(g SiO₂)⁻¹, f peaks at 68.65%, after which its growth rate slows significantly and stabilizes with further MPS addition. This behavior is primarily due to the reduced availability of hydroxyl groups as MPS dosage increases, which facilitates an increase in f . However, hydroxyl groups are spaced only a few angstroms apart, a distance relatively small compared to the size of the MPS molecules coupled onto the SiO₂ surface (Du et al., 2007). As the MPS dosage continues to rise, the substitution of hydroxyls by silane groups introduces steric hindrance, reducing the ability of additional MPS molecules to react with the SiO₂ particles. This steric hindrance limits further contact with available hydroxyls, thereby slowing the rate of increase in f (Qi et al., 2015). Moreover, only the high-activity silanol groups on SiO₂ fully react with MPS, while the inert fraction (25%–30%) remains unreacted, which indirectly validates the maximum f observed in Fig. 3d (Sneh and George, 1995). Excessive MPS concentration may also trigger self-polymerization of the modifier, forming oligomers that inhibit further SiO₂ modification. Thus, 0.5954 g·(g SiO₂)⁻¹ was determined as the optimal MPS dosage, and SiO₂ particles modified under this condition were used as reinforcing fillers for subsequent composite synthesis.

3.1.3. Dispersion stability

While carbon-carbon double bonds enhance nanoparticle-polymer matrix binding and composite performance, sufficient dispersion stability of nanoparticles is critical to ensure *in-situ* hydrogel formation in reservoirs, avoid uneven gel distribution and compromised conformance control, and maximize filler-matrix contact for improved mechanical strength and thermal stability. Based on the MLS principle, Turbiscan quantitatively characterized the colloidal stability and aggregation kinetics of 0.1 wt% SiO₂ suspensions. Fig. 4a shows the time-dependent delta transmitted intensity (ΔT) at different heights; the blue-to-red spectral shift and positive peaks at the cell top indicate nanoparticle aggregation, sedimentation, and clarified top-layer formation. To visually represent the subsidence process, a schematic was created by TURBISOFT LAB based on the raw transmission data (Fig. 4a) (Shen et al., 2022). In the diagram, dark blue corresponds to high opacity, while white represents transparency. Over time, the color of the samples shifted from dark blue to light blue, reflecting a decrease in SiO₂ content suspended at this height. Notably, MPS-SiO₂ exhibited smaller, more uniform ΔT variations than pristine SiO₂, indicating superior dispersion stability: after 3 h, pristine SiO₂ showed significant sedimentation, while MPS-SiO₂ only had slight upper-layer concentration reduction; by test end, pristine SiO₂ was nearly fully sedimented, whereas modified SiO₂ retained high suspension concentration throughout the cell. Destabilization kinetics were quantified via time-dependent global Turbiscan Stability Index (TSI) and subsidence peak thickness (δ_{peak}) (Yang et al., 2021; Joshi et al., 2022; Mekoue Nguela et al., 2024). TSI was calculated via Eq. (5), and δ_{peak} was defined as the height where ΔT reached half of ΔT_{max} (e.g., 6.22 mm for MPS-SiO₂ at 3 h). Fig. 4b shows progressive TSI increases with system destabilization; MPS-SiO₂ destabilized slower, with a 6 h TSI of 29.28, far lower than pristine SiO₂'s 51.86. Fig. 4c-derived δ_{peak} data quantified sedimentation characteristics: pristine SiO₂'s subsidence peak appeared at 26 min, with near-complete sedimentation at 278 min, a final δ_{peak} of 51.47 mm (97.11% ratio); MPS-SiO₂'s peak emerged 50 min later, with incomplete sedimentation by test end, a δ_{peak} of 16.24 mm (30.64% ratio). Moreover, sedimentation velocity (v_{sed}), calculated as the average slope of the δ_{peak} -time curve in the settlement stage, was 3.4 mm/h for MPS-SiO₂—only a quarter of that for pristine SiO₂. These phenomena are mainly attributed to the introduction of silane coupling agent, which reduces the content of silanol groups on the SiO₂ surface, weakens interparticle attraction, inhibiting agglomeration.

3.2. Morphology and structure of graphene oxide

Graphene and its derivatives are novel nanomaterials with two-dimensional planar structure and atomic thickness. Compared to chemically inert graphene, GO is more suitable as a functional filler for composite materials due to its abundant oxygen-containing groups and rich active sites. From the SEM image (Fig. 5a), GO exhibits good quasi-two-dimensionality, with its size mainly distributed between 5 and 10 μm , and shows random separation and orientation. TEM image (Fig. 5b) shows the crumbled and wrinkled nature of GO owing to the large surface area. Spectral analysis (Fig. 5c) confirms GO's rich polar groups (C–O–C, –OH, C=O, –COOH), which form hydrogen bonds with water molecules and polymer chains, endowing GO with good hydrophilicity and matrix compatibility. XRD studies show that the interlayer spacing of GO increased by 0.625 nm compared to graphite, which elevates the reactivity of its active sites. This unique structure enables full interfacial bonding of GO, thereby maximizing its contribution to the mechanical reinforcement of nanocomposites. To successfully apply GO to composite materials, it must also have good and stable dispersibility. Dispersibility tests (Table 3) show that GO's aqueous zeta potential is highly pH-dependent due to its hydroxyl and carboxyl groups. In the pH range of 1.31–9.12, the absolute zeta potential value exceeds 30 mV, verifying GO's stable dispersion capability in typical reservoir environments.

3.3. Characterization and microstructure of dual-nanocomposite hydrogel

3.3.1. Fourier transform infrared spectroscopy

FTIR spectroscopy (Fig. 6a) verified the successful fabrication of DNC hydrogels. For pure AM, the absorption peak at 3350 cm⁻¹ is attributed to the symmetric stretching vibration of N–H, while the peak at 3184 cm⁻¹ corresponds to its asymmetric stretching vibration. The presence of amide groups (–CONH₂) within AM is evidenced by the peak at 1674 cm⁻¹, and the bending vibration of the –CH₂ group is indicated by a peak at 1429 cm⁻¹. Additionally, the characteristic absorption peak at 1612 cm⁻¹ corresponds to the C=C stretching vibration in AM (Li et al., 2023). This peak is notably absent in the spectra of the synthesized gels, indicating that AM fully reacts with the comonomers during hydrogel formation. In the spectra of the hydrogels, the peaks situated around 1725 cm⁻¹ and 1656 cm⁻¹ originate from –C=O vibrations of –COO⁻ and –CONH₂, respectively (Wang et al., 2024). A peak at 1454 cm⁻¹ corresponds to the bending vibration of C–H (–CH₂–), while a peak at 1417 cm⁻¹ arises from the C–H stretching of the –N⁺(CH₃)₂ group (Devi et al., 2024). Additionally, characteristic absorption peaks at 1176 cm⁻¹ and 1036 cm⁻¹ confirm the presence of sulfonate groups (Li et al., 2020). The peak at 1123 cm⁻¹ is related to the stretching of C–N. The detection of these monomer-derived functional groups confirmed the full incorporation of all monomers into the hydrogel network. Additionally, a new peak at 473 cm⁻¹ (Si–O–Si bending vibration) verified the successful encapsulation of MPS-SiO₂ within the NC and DNC hydrogels.

3.3.2. Nuclear magnetic resonance hydrogen spectrum

NMR spectroscopy was utilized to verify the successful incorporation of GO nanosheets into DNC hydrogels. Fig. 6b shows ¹H NMR spectra of PAM and PAM/GO in DMSO. For pure PAM, the absorption peaks at 1.45 and 2.04 ppm correspond to protons in –CH₂ and –CH groups, respectively. Peaks at 6.81 and 7.13 ppm represent protons associated with R–CO–NH₂ groups, confirming the presence of primary functional groups in the PAM structure. In the PAM/GO spectrum, the characteristic peaks of PAM exhibit a distinct high-field shift (lower chemical shift), indicative of PAM backbone structural modification induced by GO interactions. Additionally, new peaks with distinct shapes appear around 6.90 ppm, indicating hydrogen-bonding interactions (COOH...NH₂), which confirm how GO interacts with PAM to form hydrogen bonds. The above phenomena prove that GO nanosheets

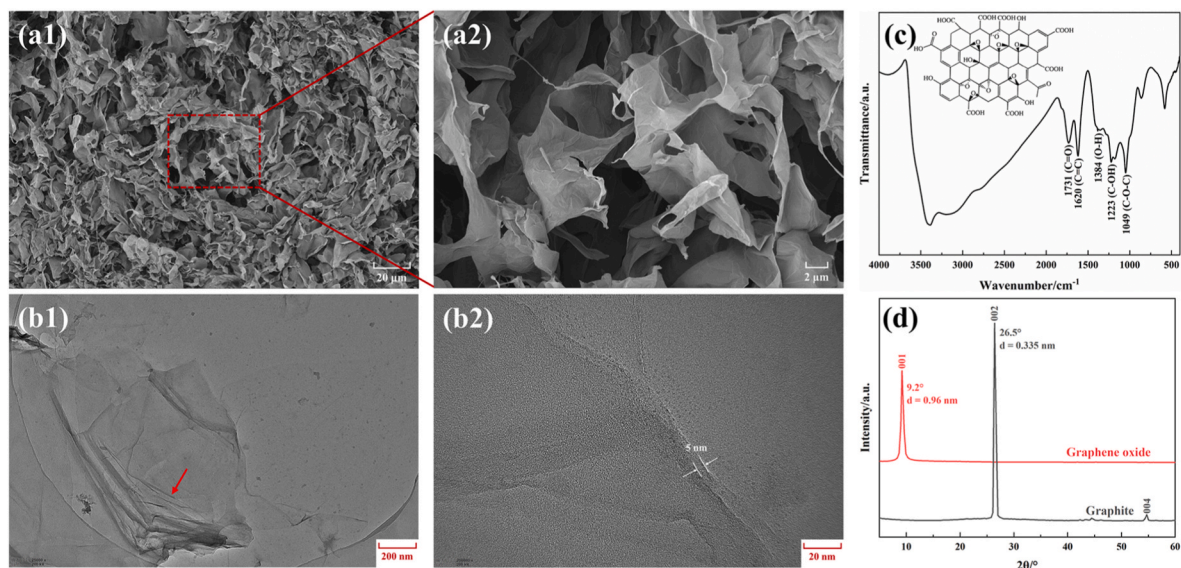


Fig. 5. Morphology and characterization of GO. (a) SEM images, (b) TEM images, (c) FTIR spectrum, and (d) XRD patterns.

Table 3

Zeta potential of GO dispersion at 25 °C for different pHs.

pH	0.31	1.31	1.79	3.02	5.51	9.12
ζ /mV	-9.84	-34.05	-35.97	-50.18	-47.11	-34.96

participate in the construction of the gel skeleton through physical bonding (Haruna et al., 2019; Qin et al., 2023).

3.3.3. Scanning electron microscopy and energy dispersive spectroscopy

The successful synthesis of DNC hydrogels was further affirmed by morphological and elemental characterization. As shown in Fig. 7a, digital photographs demonstrate a progressive color deepening with nanoparticle incorporation: the base hydrogel is uniform and transparent, NC hydrogel is turbid, and DNC hydrogel shows a distinct black color. Lyophilized SEM images reveal a honeycomb-like internal microstructure across all samples, which enlarges specific surface area and improves water absorption/retention capacity. Notably, nanoparticle addition (MPS-SiO₂ and GO) drastically reduces mean pore size (base hydrogel: 10.87 μ m; NC hydrogel: 5.68 μ m; DNC hydrogel: 4.83 μ m) and optimizes pore size uniformity. This occurs because MPS-SiO₂ and GO serve as crosslinking sites, increasing crosslinking density and limiting the expansion of the gel's microporous matrix, thereby

creating a more robust and stable network structure (Yang et al., 2018; Zhang et al., 2019; Almeida et al., 2024; Deng et al., 2025). However, the distribution and reinforcement patterns of the two nanomaterials within the gel matrix differ significantly. Observations of the hydrogel cross-sections reveal that the base hydrogel has a smooth, flat surface (blue box), while numerous nanoscale spherical particles (~20 nm in diameter) are visible within the hydrogel framework (green box), indicating the encapsulation of MPS-SiO₂ within the matrix. Additionally, GO exhibits a “reinforcing rib”-like effect within the hydrogel (red box), bridging the polymer skeletons. This variance primarily arises from the distinct cross-linking mechanisms of the two nanomaterials. Nanosized MPS-SiO₂ nanoparticles act as multifunctional crosslinkers through covalent silane bridges and hydrogen bonding, owing to their high surface energy and polymerizable reactive groups. This allows them to function as reaction centers and crosslinking sites within the polymer matrix, actively contributing to the construction of the gel network (Ji et al., 2017; Tang et al., 2020). In contrast, GO serves as a physical filler and crosslinker, providing bridging effects and mechanical interlocking due to its high aspect ratio and size effect (Long et al., 2022; Qin et al., 2023). Together, these interactions synergistically enhance the overall performance of DNC hydrogel. EDS analysis and elemental mapping (Fig. 7b and c) confirm the presence of C, N, O, Na, and S in descending content order, with uniform distribution and no obvious agglomeration. C, N, O account for >95% of total atoms (constituting the gel matrix backbone),

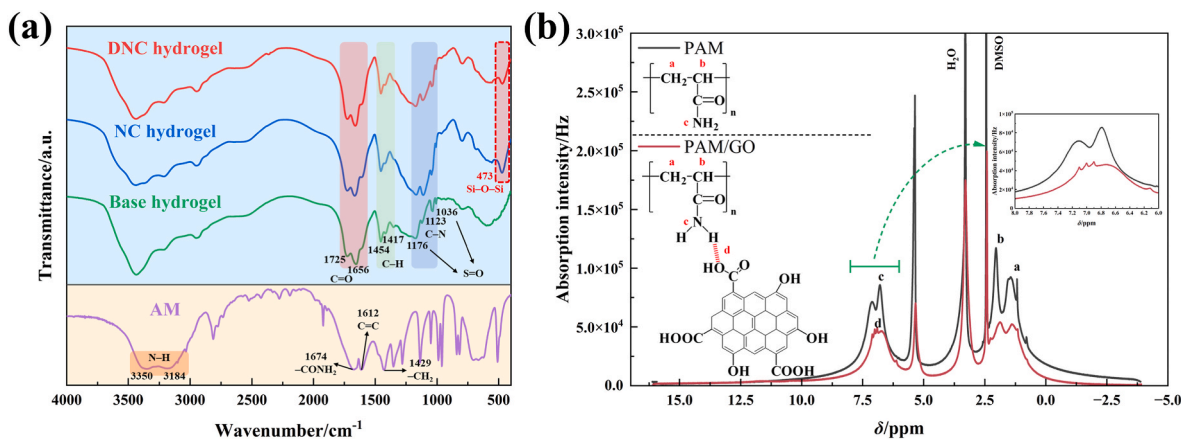


Fig. 6. Characterization by (a) FTIR spectra and (b) ¹H NMR spectra.

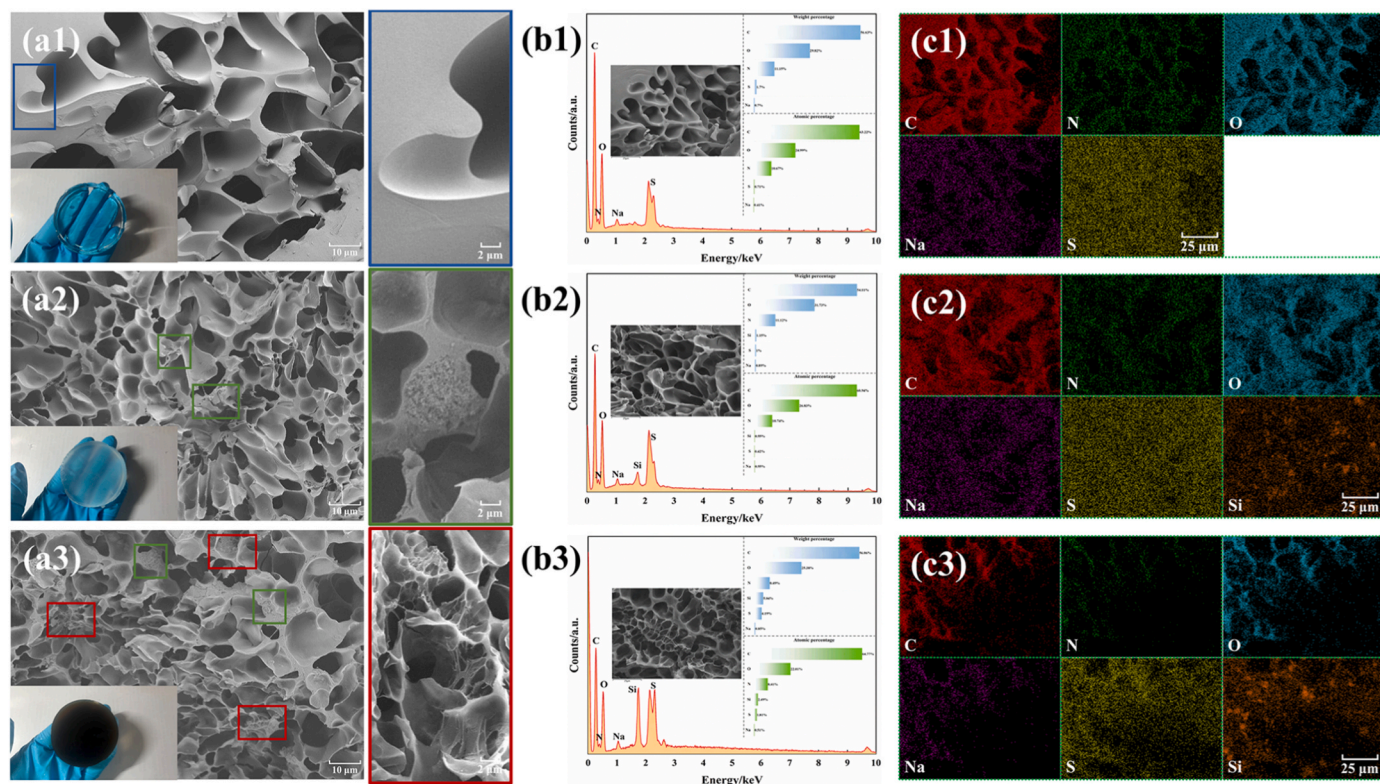


Fig. 7. Morphology and elemental characterization. (a) Digital photographs and SEM images, (b) EDS spectra and elemental chemical composition, and (c) corresponding EDS mappings of (1) base hydrogel, (2) NC hydrogel, and (3) DNC hydrogel.

while Na and S derive from functional monomers (DMAPS, SSS). Si is detected in both NC and DNC hydrogels, which is distributed at network nodes as crosslinking sites—consistent with SEM observations. Notably, DNC hydrogel has a significantly higher Si atomic percentage (2.49%) than NC hydrogel (0.55%). This is primarily because the GO surface contains numerous oxygen-containing functional groups, which form hydrogen bonds or electrostatic interactions with the silicon-oxygen bonds of MPS-SiO₂. Such interactions can promote changes in the distribution of silica particles within the gel matrix and even encourage the aggregation of silica near GO, resulting in a composite enhancement effect.

The results confirm the specific mechanism underlying the formation of the DNC hydrogel, as illustrated in Fig. 1. Polymer chains cross-linked by MBA and polymeric monomers (e.g., AM and AA) form a robust backbone that supports the honeycomb-like structure. MPS-SiO₂ serve as multifunctional crosslinking sites, enhancing crosslinking density and stabilizing the three-dimensional network. Concurrently, GO strengthens the gel matrix through its rigid two-dimensional structure, capacity to establish intermolecular hydrogen bonds, and promotion of a more uniform silica distribution, thereby improving mechanical strength and energy dissipation. These synergistic effects from multiple cross-linked structures collectively enhance the comprehensive performance of the DNC hydrogel.

3.4. Effect of nanofiller on gelling properties and thermal stability

3.4.1. Gelling properties

Key evaluation criteria for conformance control agents focus on injectivity, gelation time, gel strength, and thermal stability, summarized as: injectable, targetable, blockable, and durable. This section investigates the synergistic enhancement of gel properties by MPS-SiO₂ and GO, as well as their distinct functional roles. Fig. 8a compares the initial viscosity of various gelant solutions to that of distilled water. The viscosity of the base hydrogel solution is nearly equivalent to that of

water. The suspended nano-SiO₂ particles restrict fluid movement, resulting in a slight increase in viscosity. For the DNC hydrogel system, the viscosity increases markedly to 14.9 mPa s. This increase is primarily attributed to the lamellar structure of GO, which enhances solution resistance through physical filling. Additionally, stacking and bridging interactions among GO sheets, between GO and SiO₂, and slight cross-linking with polymer monomers via hydrogen bonding, contribute to this effect. However, these *in-situ* crosslinked DNC hydrogels still maintain a relatively low initial viscosity, enabling smooth injection into subsurface formations and preventing shear-induced degradation, often seen in surface-crosslinked gels.

Fig. 8b and c illustrate the storage modulus G' and loss modulus G'' evolution during isothermal gelation of hydrogels with varied monomer contents, with signals clarified via locally weighted scatterplot smoothing (LOWESS). Key gelation parameters are summarized in Table 4. Viscoelastic profiles reveal three distinct gelation stages: induction, growth, and maturation. The transitions between these stages are characterized by two critical time points: the initial gelation time (t_0), marking the onset of rapid G' growth, and the gelation completion time (t_c), signifying the completion of the gel network. Notably, the gelation completion time t_c is defined as the point at which the rate of increase in G' between two adjacent data points falls below 0.1%, indicating the full formation of the gel structure (Guo et al., 2022; Tang et al., 2024). Prior to gelation, the modulus values of each hydrogel solution remain low, with significant fluctuations in the measured values. Notably, only the DNC hydrogel system exhibits a discernible G' signal, allowing continuous monitoring owing to the preliminary structural support provided by MPS-SiO₂ and GO. Fig. 8b demonstrates that the addition of MPS-SiO₂ significantly accelerates gelation by introducing extra chemical crosslinking sites. The t_0 was advanced from 215 min to 198 min, and the t_c was advanced from 573 min to 540 min. Intriguingly, the incorporation of GO delays the gelation process of the DNC hydrogel. Compared with the base hydrogel, the t_0 reached 250 min, extended by 16.3%, and the t_c reached 610 min, extended by

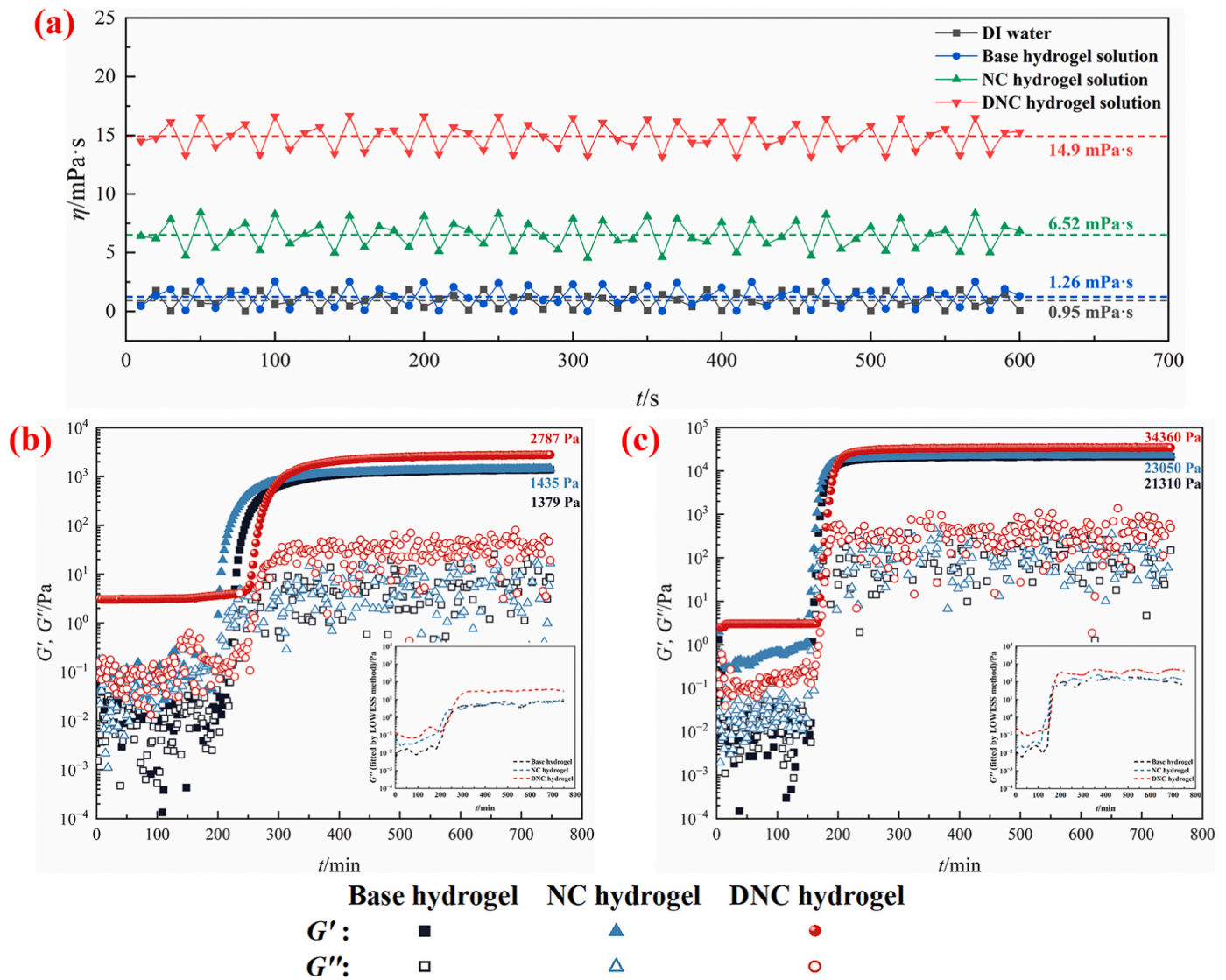


Fig. 8. Gelation kinetics of hydrogels. (a) Comparison of initial viscosity of different gelant solutions with distilled water. Changes in storage modulus and loss modulus during isothermal gelling process for gels with (b) 2.5 wt% and (c) 5 wt% AM contents.

Table 4

Gelling characteristic parameters for hydrogels.

Hydrogel type		t_0 /min	t_c /min	G'/Pa	G''/Pa
Base hydrogel	(with 2.5 wt% AM content)	215	573	1379	7.93
NC hydrogel		198	540	1435	9.97
DNC hydrogel		250	610	2787	30.04
Base hydrogel	(with 5.0 wt% AM content)	155	348	21310	64.32
NC hydrogel		150	338	23050	129.26
DNC hydrogel		168	390	34360	405.25

6.5%. This delay provides extended operational time for on-site construction personnel, ensuring that the hydrogel solution reaches the designated target zone effectively. Similar phenomena can also be observed in Fig. 8c. The delayed gelation effect primarily arises from the hydrogen bonding interactions between GO and the crosslinker MBA. As verified by the molecular dynamics (MD) simulations, GO with micron-scale lateral dimensions can adsorb and encapsulate MBA via hydrogen bonding, effectively converting part of the MBA into an

inactive associated state and reducing the concentration of effectively available crosslinkers at the early gelation stage (Fig. 9). According to the MD results and supported by Huang et al. (2018), MBA in the solution exists in three dynamically equilibrated states: free, mono-complex, and di-complex. Only free MBA participates in the gel network construction; as free MBA is gradually consumed, the dynamic equilibrium shifts, and the bound MBA is slowly released into the solution. This “slow-release” mechanism regulated by GO effectively prolongs the gelation time of the hydrogel.

Post-gelation modulus analysis shows that all gels are dominated by a high-elastic state, with G' two to three orders of magnitude higher than G'' . With increasing AM content from 2.5 wt% to 5.0 wt%, G' increases tenfold. Mono-incorporation of MPS-SiO₂ contributes marginally to gel strength, with G' increasing by less than 10%. Although the inclusion of MPS-SiO₂ enhances the gel system by forming a denser three-dimensional network structure, as described in Sec. 3.3, NC hydrogel remains primarily reliant on organic polymer chains. Due to its nanoscale dimensions, MPS-SiO₂ mainly serves as an active crosslinking site within the network, limiting its ability to deliver a rigid response to external mechanical forces. Consequently, its contribution to the G' of the gel system is relatively modest. Upon further incorporation of GO, the strength of hydrogels with two different AM contents increased

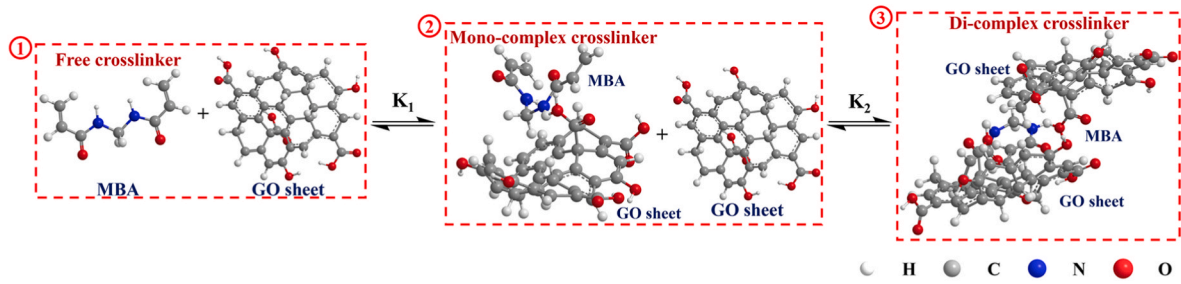


Fig. 9. Dynamic equilibrium of MBA in different states in gelant solution.

significantly. Specifically, G' rose from 1379 and 21310 Pa to 2787 and 34360 Pa, corresponding to increases of 102% and 61%, respectively. The inclusion of GO markedly enhances the gel's energy storage capacity and mechanical strength during the maturation phase. This enhancement arises from the reinforcing properties of its two-dimensional layered structure, synergistic physicochemical interactions, elevated crosslinking density, and the optimization of anisotropic networks (Zhang et al., 2019; Narimani et al., 2023; Karchoubi et al., 2024). Mechanically, the polymeric network of DNC hydrogel is strengthened through a combination of covalent and noncovalent interactions. Covalent bonds act as permanent crosslinking sites, maintaining structural integrity, while noncovalent bonds serve as sacrificial sites, dissipating energy upon rupture to prevent catastrophic failure (Fukao et al., 2020).

Moreover, the well-dispersed rigid GO nanosheets reduce stress concentration and inhibit crack initiation or propagation, effectively enhancing the gel's durability. This multifaceted enhancement mechanism provides valuable theoretical insights and practical guidance for the development of high-performance gel materials. Notably, the above discussion focuses on nanofiller-regulated gelation characteristics. For practical oilfield applications, gelation time and strength can be further optimized via cold water injection, precise formulation adjustments, and the incorporation of retarders.

Mechanical rheometers primarily measure macroscopic viscoelastic responses but are incapable of capturing micro-scale viscoelastic changes within the sample. Consequently, the microrheological response was monitored *in situ* to investigate the microstructural

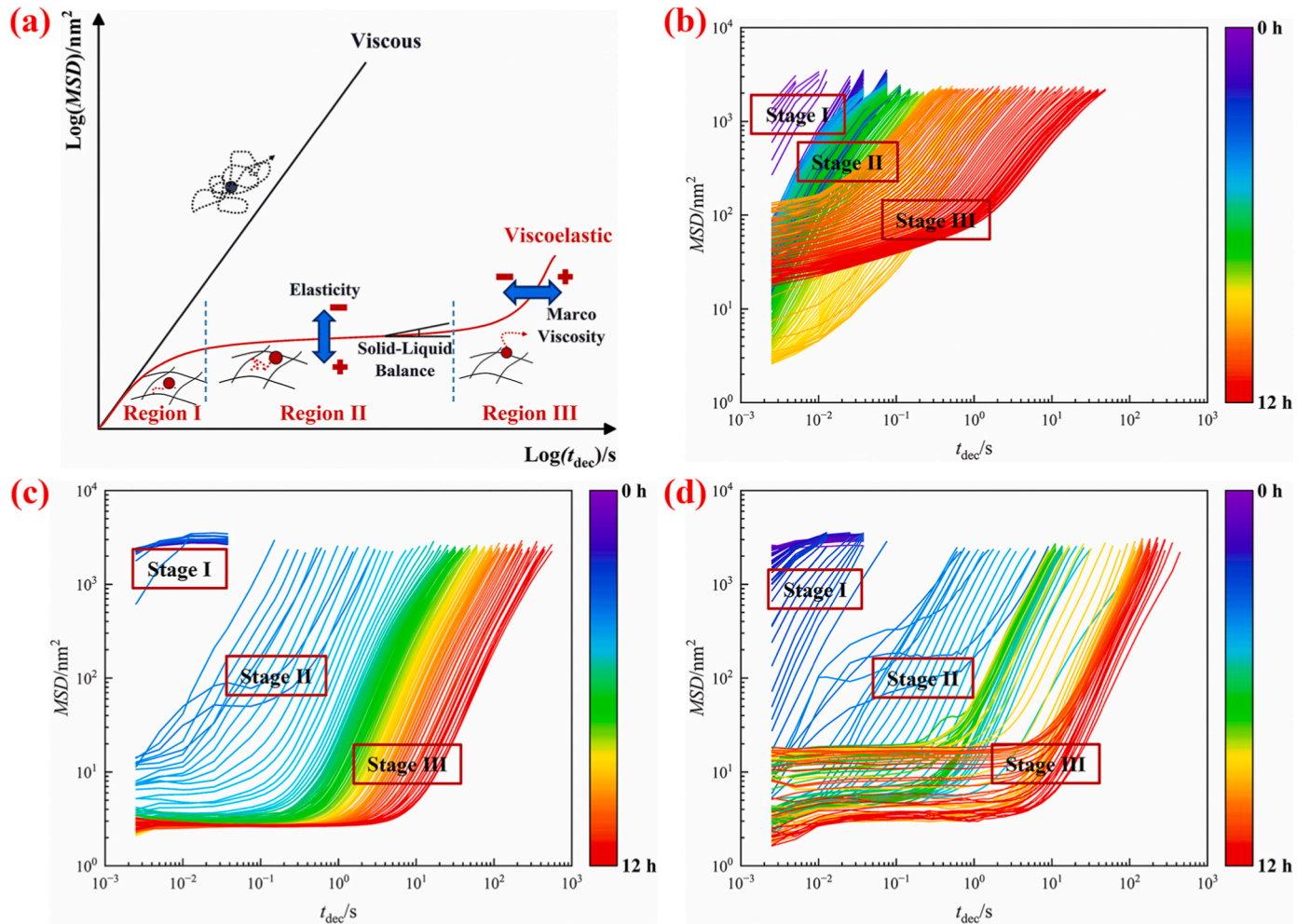


Fig. 10. Microrheological properties in gelling process. (a) MSD curves of purely viscous and viscoelastic materials. MSD curves at different hydration times for (b) base hydrogel, (c) NC hydrogel, and (d) DNC hydrogel within 12 h at 60 °C (2.5 wt% AM content).

evolution of the DNC hydrogel by tracking the Brownian motion of probe particles. From an optical microrheology perspective, the *MSD* curve exhibits linear growth with t_{dec} for Newtonian fluids. In contrast, for viscoelastic materials, the curve becomes nonlinear, indicating viscoelastic behavior (Fig. 10a) (Long et al., 2022). With respect to a viscoelastic sample, the *MSD* curve is generally divided into three distinct regions: (I) a nearly linear region at very short t_{dec} , representing unconstrained probe particle motion; (II) a plateau region at intermediate t_{dec} , where particle motion is restricted by neighboring particles and the curing medium; and (III) another linear region at long t_{dec} , where particles escape from the “cages” formed by their surroundings. The *MSD* curves enable the extraction of several key parameters that provide deeper insights into the microrheological characteristics of the specimens (Tisserand et al., 2012; Wu et al., 2017):

- Elasticity index (*EI*), derived as the inverse of the *MSD* plateau value, quantifies the elasticity of the specimen. A lower elastic plateau corresponds to greater elasticity.
- Macroscopic viscosity index (*MVI*), defined as the inverse slope of *MSD* curve in the later linear scale on a double-linear scale, corresponds to a viscosity index at zero shear rate.
- Solid-liquid balance (*SLB*), corresponding to the slope of the *MSD* plateau, characterizes the ratio between solid- ($SLB = 0$) and liquid-like ($SLB = 1$) behaviors of the specimen.

Fig. 10b–d displays the *MSD* curves of the hydrogels at various hydration times over a 12-h period at 60 °C, with curve colors matching the corresponding test times. During the gelling process, the downward and rightward shifts of the *MSD* curves indicate increasing viscoelasticity and the progressive development of gel networks over time. Analogous to macroscopic rheological curves, the *MSD* curves also exhibit three distinct stages over time, corresponding to the gel formation process. In Stage I, the monomers remain uncrosslinked, and the gelant solution exhibits Newtonian fluid behavior. The *MSD* and t_{dec} display an approximately linear relationship. In Stage II, the *MSD* curves progressively behave an inclined, staircase-like pattern, with significant intervals between curves. This behavior indicates the initial construction of the gel network system and a rapid increase in viscoelasticity. In Stage III, the downward and rightward progression of the curves slows, with curves eventually overlapping. At this point, the statistical movement patterns of the probe particles remain unchanged, indicating that the gel network structure has fully matured. A comparison of the *MSD* curve trends for the three types of gels reveals that changes in the *MSD* curve for NC hydrogel become negligible during the mid-to-late testing period. Notably, the plateau region (Region II) in Fig. 10c shows almost complete overlap, indicating the fastest gelation rate among the three samples. In contrast, DNC hydrogel exhibits significant changes in its *MSD* curve even in the later stages. Microrheometric data indicate gelation times of 602, 549, and 648 min for the three gels, respectively. These findings further highlight the strong crosslinking effect of MPS-SiO₂ and the “slow-release” behavior of GO during the gelation process. Key microrheological parameters are summarized in Table 5. Compared to the base hydrogel, DNC hydrogel demonstrates substantially elevated *EI* and *MVI* values alongside a notably reduced *SLB*, categorizing it as a rigid hydrogel with superior viscoelastic characteristics. These findings strongly agree with the data derived from the oscillatory rheometer, reinforcing the reliability of the analysis.

Table 5
Microrheological characteristic parameters for hydrogels.

Hydrogel type	<i>EI</i> /nm ⁻²	<i>MVI</i> /(nm ⁻² ·s)	<i>SLB</i>
Base hydrogel	0.0246	0.0212	0.2551
NC hydrogel	0.3277	0.2485	0.0617
DNC hydrogel	0.3702	0.3894	0.0656

3.4.2. Thermal stability

Thermal stability is a critical factor affecting the time required for hydrogels to establish and maintain long-term effective plugging in reservoirs. The performance of hydrogels with varying AM contents was evaluated over the aging period, with representative photographs presented in Fig. 11a for 1, 60, and 180 days. Upon gelation completion and the start of thermal stability testing (Day 1), all gels showed exceptional strength, with GSC ratings of H or higher. After 60 days of aging, gel strength decreased to varying degrees, with lower monomer content correlating to more significant declines (strength range: B to I grade). However, a minimum GSC rating of D is required for reservoir conformance control and water shutoff applications (Zhu, 2018). This indicates that hydrogels with low monomer content fail to achieve long-term effective plugging in high-temperature reservoirs. After 180 days of testing, the base hydrogel requires an AM content of at least 5.0 wt% to maintain functionality, whereas the composite hydrogel achieves long-term thermal stability with a monomer content as low as 2.5 wt%. Furthermore, hydrogels incorporating both SiO₂ and GO exhibits superior thermal stability compared to those containing only SiO₂. At a lower AM content (2.5 wt%), the gel surface of DNC hydrogel exhibits only slight deformation under inclined conditions (Fig. 11b), demonstrating its capability to sustain high-strength plugging within the reservoir.

The superior thermal stability of DNC hydrogel is elucidated through the gel microstructure depicted in Fig. 7a and the schematic in Fig. 12. In the base hydrogel, the microporous structure composed of polymer chains is prone to collapse and relatively sparse, allowing water molecules to escape from the gel interior with minimal resistance at elevated temperatures (Fig. 12a). In NC hydrogel, functionalized silica nanoparticles are embedded within the gel matrix, enhancing the crosslinking density of the network. The resulting stronger and denser 3D network structures delay collapse, help the gel system retain water, minimize water escape pathways, and enhance the thermal stability of the material (Fig. 12b). Moreover, in DNC hydrogel, GO further augments the crosslinking density of the gel structure, thereby effectively narrowing fluid flow channels. Most importantly, its unique two-dimensional structure allows it to serve as a “water-blocking baffle” within the gel matrix, boosting water impermeability (Hackman and Hollaway, 2006; Mohammed et al., 2015; Qin et al., 2023). Water molecules are forced to navigate around these sheets when escaping, creating intricate, tortuous diffusion paths (Fig. 12c). This process significantly restricts water mobility, thereby introducing a novel anti-thermal mechanism.

Since the GSC method provides only a qualitative assessment of long-term thermal stability, it cannot accurately capture the variation in gel strength over time. Therefore, the *MVI* variations of the samples were continuously monitored using the microrheometer (Fig. 13). The *MVI* evolution exhibits an initial stabilization phase followed by a subsequent decline. For DNC hydrogels, the downward trend is markedly decelerated, indicating that increased microstructural complexity prolongs the stabilization period. Fig. 13 also marks the macroscopic destabilization time determined via visual observation. Destabilization phenomena are detected at the microscopic scale well before observable macroscopic changes, indicating that microstructural instability initiates prior to noticeable changes at the macroscopic level.

Overall, MPS-SiO₂ and GO exhibit both overlapping and distinct functionalities, enabling a synergistic effect that significantly enhances the gelation characteristics and thermal stability of the DNC hydrogel.

3.5. EOR potential of dual-nanocomposite hydrogel

In this part, the breakthrough pressure difference (ΔP_{max}), plugging rate (η), and residual resistance factor (F_{rr}) were employed to evaluate the plugging efficiency of the hydrogels in one-dimensional core flow experiments. After gel treatment, higher characteristic parameters indicated that the gel could effectively divert flow to low-permeability

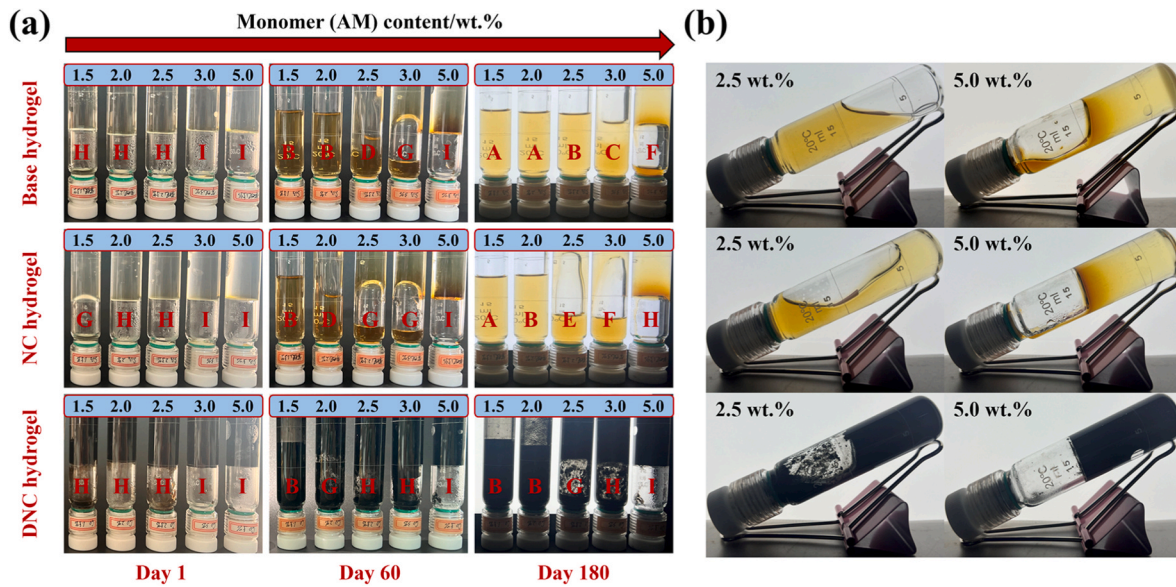


Fig. 11. Thermal stability of three kinds of hydrogels aged at 80 °C during 180 days. (a) Comparison of the gel strength at different aging time, and (b) flow state of gels with 2.5 wt.% and 5.0 wt.% AM contents after 180 days.

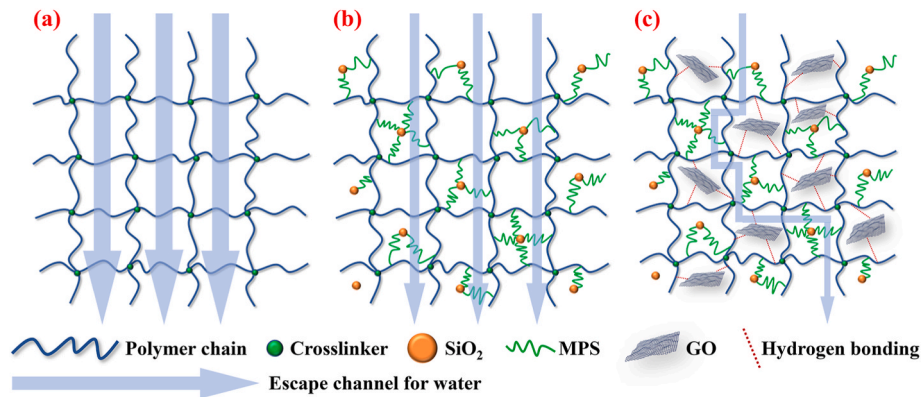


Fig. 12. Escape channels for water in different hydrogels. (a) Base hydrogel, (b) NC hydrogel, and (c) DNC hydrogel.

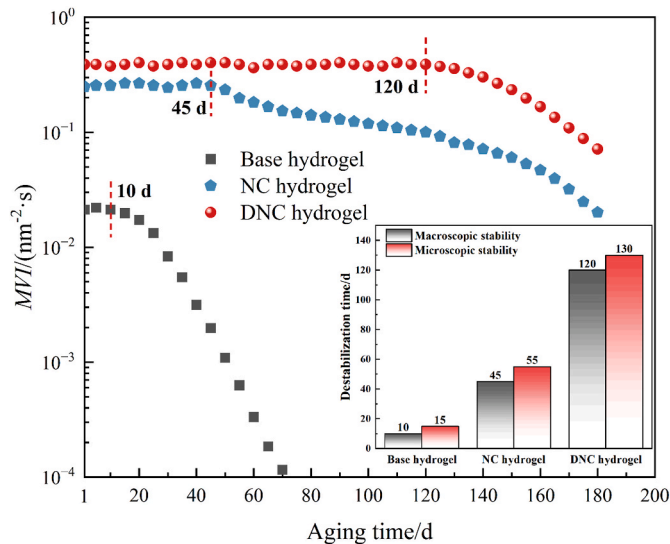


Fig. 13. MVI versus aging time for three hydrogels with 2.5 wt.% AM content.

zones. As listed in Table 6, DNC hydrogel exhibited a notable $\Delta P_{\max}$ of 0.207 MPa in the core with an initial water permeability of $451.2 \times 10^{-3} \mu\text{m}^2$. The calculated F_{rr} was 22.67 based on the permeability before and after gel treatment, indicating a diversion of 95.59% of the brine flow, as derived from the plugging rate equation. This gel demonstrated superior plugging efficiency compared to the base hydrogel (η 76.80%, F_{rr} 4.31) and NC hydrogel (η 87.61%, F_{rr} 8.07) under similar conditions. On the one hand, the incorporation of nano-fillers significantly enhances the mechanical strength of the system. On the other hand, as the pressure difference across the pore-throats increases under water injection pressure, weaker noncovalent bonds rupture preferentially over covalent bonds, inducing large-scale elastic deformation. This deformation promotes energy dissipation, mitigates stress concentration, and ultimately increases the $\Delta P_{\max}$ of the hydrogel

Table 6

Plugging capacity of hydrogels in one-dimensional cores.

Hydrogel type	$\Delta P_{\max}/\text{MPa}$	$k/\times 10^{-3} \mu\text{m}^2$		$\eta/\%$	F_{rr}
		k_0	k'		
Base hydrogel	0.050	451.2	104.7	76.80	4.31
NC hydrogel	0.093	458.4	56.8	87.61	8.07
DNC hydrogel	0.207	458.0	20.2	95.59	22.67

in porous media. Notably, unlike commercial particle gels, the DNC hydrogel did not exhibit extrusion under long-term steady flow conditions, indicating its potential for permanent, effective plugging of target channeling zones. Fig. 14a further illustrates the η and F_{rr} of DNC hydrogel in cores with varying permeabilities, exceeding 85% and 7.5, respectively. These results demonstrated that DNC hydrogel met the requirements for effectively plugging the dominant seepage channels.

Building on the excellent physicochemical properties and superior plugging performance of DNC hydrogel, visualization tests and core flooding experiments were conducted to assess its effectiveness in conformance control and EOR. Fig. 14b illustrates the oil distribution characteristics in the visualization model after crude oil saturation and various displacement strategies. Darker colors represent areas with higher oil content. In the high-permeability zone, glass beads are arranged in a single layer due to their size matching the cavity thickness, resulting in higher light transmission. However, this does not imply lower initial oil saturation compared to medium- and low-permeability zones. Owing to its limited conformance control capability, the surfactant solution predominantly channels through high-permeability pathways between injection and production wells, leading to poor sweep efficiency in medium- and low-permeability zones, where the overall sweep efficiency is only 26.25%. In contrast, the SP system improves sweep efficiency by reducing the water-oil mobility ratio, achieving an overall sweep efficiency of 66.09% in medium- and low-permeability zones.

However, significant areas of ineffective or inefficient sweep remain, particularly in the low-permeability layers, along both sides of the producer–injector connection line. The pre-injection of 0.15 PV DNC hydrogel before SP flooding effectively mitigates this limitation. The gelant solution, characterized by its initially low viscosity, preferentially infiltrates high-permeability channels and forms robust plugs after gelation (as indicated by the green dashed circles in Fig. 14b). As a result, the SP flooding fluid is redirected toward medium- and low-permeability zones, significantly increasing the sweep range. With this injection strategy, the final overall sweep efficiency in medium- and low-permeability layers reached an impressive 92.72%, representing a 40.29% improvement in profile modification compared to SP flooding alone. This highlights the remarkable conformance control capability of DNC hydrogel.

To better evaluate the EOR potential of DNC hydrogel in heterogeneous reservoirs, two sets of oil displacement experiments were conducted, as illustrated in Fig. 14c and d. Under the experimental conditions, the oil recovery during water flooding stage was approximately 24%. For the SP system alone, the maximum injection pressure difference reached 969 kPa, the minimum water cut dropped to 67.75%, and the ultimate oil recovery was 43.03%—a 19.26-percentage-point increase relative to water flooding. When 0.15 PV of gelant solution was injected prior, the pressure difference curve showed no significant increase, indicating that the solution successfully entered the high-

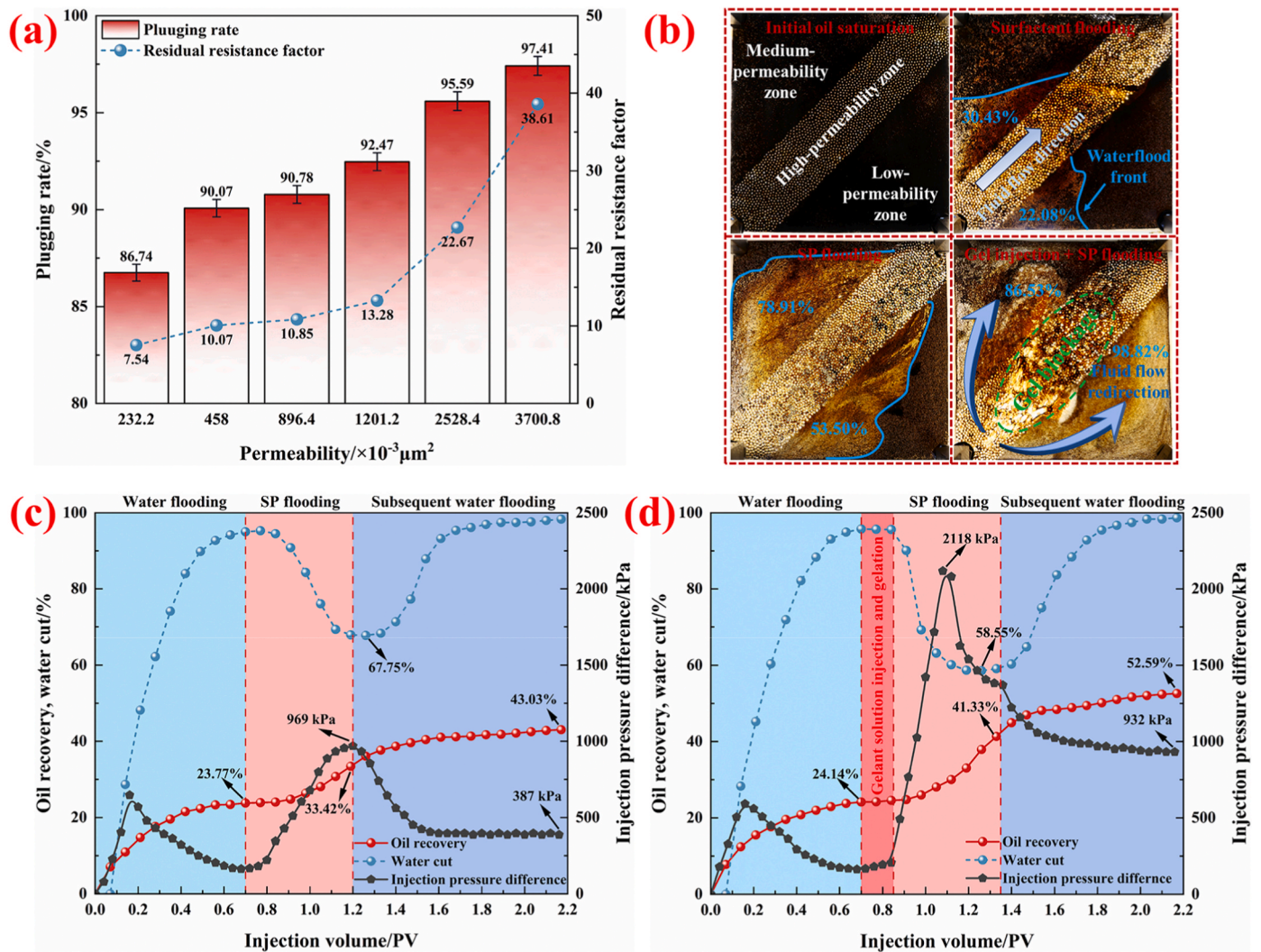


Fig. 14. Conformance control and EOR capacity of DNC hydrogel. (a) Plugging performance in cores with varying permeabilities; (b) conformance control tests by visualized model; and dynamic production curves under different strategies: (c) 0.5 PV SP system and (d) 0.15 PV gel + 0.5 PV SP system.

permeability layer along the water flow channel. Post-gelation, SP system injection rapidly increased the pressure difference to 2118 kPa, a 119% increase compared to the SP system alone. Driven by the significant pressure difference, the subsequent fluid flow was redirected to low-permeability layers, greatly enhancing oil recovery. This injection strategy reduced the minimum water cut to 58.55% and raised the ultimate oil recovery to 52.59%—an additional 9.56-percentage-point improvement over SP flooding alone. During the subsequent water flooding stage, the stable pressure difference remained at 932 kPa, 1.41 times higher than that of the non-gel injection group, providing greater potential for further conformance control after chemical flooding.

These results fully verify the efficacy of DNC hydrogel in conformance control and EOR, laying a solid foundation for the development of high-performance oilfield materials.

4. Conclusions

This study synthesized a dual-nanocomposite (DNC) hydrogel integrating MPS-SiO₂ and GO into a multi-crosslinked polymer network, with its superior performance verified by systematic experiments and quantitative analyses. Key conclusions are as follows:

1. Silane modification endows MPS-SiO₂ with good matrix compatibility and dispersion stability; GO's two-dimensional structure and oxygen-containing groups provide bonding sites and physical reinforcement for the hydrogel.
2. DNC hydrogel was prepared via *in-situ* free radical polymerization, forming a stable 3D network through covalent (crosslinker/silane bridges) and non-covalent (hydrogen bonding) crosslinking, which enhances mechanical strength and energy dissipation.
3. The hydrogel exhibits excellent gelling properties (extended gelation time, high viscoelasticity) for effective reservoir deployment, and maintains structural integrity and high strength after 180 days of aging at elevated temperatures, attributed to MPS-SiO₂/GO synergy that restricts water mobility and resists thermal collapse.
4. Core flooding experiments confirm the DNC hydrogel achieves a 95.59% plugging rate and 22.67 residual resistance factor, outperforming conventional and NC hydrogels; it redirects fluid flow into low-permeability layers, boosting oil recovery by 9.56 percentage points versus the non-gel control.

In summary, DNC hydrogel holds great potential for hydrocarbon recovery, and the nanomaterial-polymer synergy provides a promising strategy for developing advanced materials tailored to challenging reservoir conditions.

CRediT authorship contribution statement

Xuechen Tang: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Resources, Visualization, Writing – original draft, Writing – review & editing. **Xinji Du:** Data curation, Formal analysis, Investigation, Writing – original draft. **Yiqiang Li:** Funding acquisition, Project administration, Resources, Supervision, Validation. **Tao Song:** Formal analysis, Validation. **Jinxin Cao:** Software, Visualization. **Zheyu Liu:** Data curation, Investigation. **Shilin Zeng:** Investigation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors thank the National Natural Science Foundation of China

(Grant U23B6003) for the support of this work.

Data availability

Data will be made available on request.

References

- Almeida, A.I.A.D., Carvalho, L.D.O., Lopes, R.C.F.G., et al., 2024. Enhanced polyacrylamide polymer hydrogels using nanomaterials for water shutoff: morphology, thermal and rheological investigations at high temperatures and salinity. *J. Mol. Liq.* 405, 125041. <https://doi.org/10.1016/j.molliq.2024.125041>.
- Bai, B., Zhou, J., Yin, M., 2015. A comprehensive review of polyacrylamide polymer gels for conformance control. *Petrol. Explor. Dev.* 42, 525–532. [https://doi.org/10.1016/S1876-3804\(15\)30045-8](https://doi.org/10.1016/S1876-3804(15)30045-8).
- Dai, C., You, Q., Zhao, M., et al., 2023. *Principles of Enhanced Oil Recovery*. Springer Nature.
- Dems, D., Freeman, R., Riker, K.D., et al., 2021. Multivalent clustering of adhesion ligands in nanofiber-nanoparticle composites. *Acta Biomater.* 119, 303–311. <https://doi.org/10.1016/j.actbio.2020.11.009>.
- Deng, J., Zhao, H., Zheng, H., et al., 2025. CO₂-responsive preformed particle gels with high strength for CO₂ conformance control in heterogeneous reservoirs. *Fuel* 379, 133040. <https://doi.org/10.1016/j.fuel.2024.133040>.
- Devi, K., Sharma, A., Kumar, R., et al., 2024. Designing of tragacanth gum-poly(2-(methacryloyloxy) ethyl] dimethyl-(3-sulfopropyl) ammonium hydroxide) and poly (acrylamide) based hydrogels for drug delivery and wound dressing applications. *Med. Novel Technol. Devices* 22, 100303. <https://doi.org/10.1016/j.mednrd.2024.100303>.
- Doan, V.T.C., Dao, T.M., Huynh, T.A., et al., 2024. A simple and efficient synthesis of 5-hydroxymethylfurfural from carbohydrates using acidic ionic liquid grafted on silica gel. *RSC Adv.* 14, 17480–17490. <https://doi.org/10.1039/D4RA02487G>.
- Du, J., She, X., Zhu, W., et al., 2019a. Tough hybrid hydrogels based on simultaneous dual sol-gel technique and radical polymerization. *J. Appl. Polym. Sci.* 136, 47742. <https://doi.org/10.1002/app.47742>.
- Du, J., She, X., Zhu, W., et al., 2019b. Super-tough, anti-fatigue, self-healable, anti-fogging, and UV shielding hybrid hydrogel prepared via simultaneous dual in situ sol-gel technique and radical polymerization. *J. Mater. Chem. B* 7, 7162–7175. <https://doi.org/10.1039/C9TB01625B>.
- Du, J., Xu, S., Feng, S., et al., 2016. Tough dual nanocomposite hydrogels with inorganic hybrid crosslinking. *Soft Matter* 12, 1649–1654. <https://doi.org/10.1039/C5SM02790J>.
- Du, L., Xiao, Y., Jiang, Z., et al., 2025. A high temperature-resistant, strong, and self-healing double-network hydrogel for profile control in oil recovery. *J. Colloid Interface Sci.* 679, 490–502. <https://doi.org/10.1016/j.jcis.2024.10.077>.
- Du, Y., Du, X., George, S.M., 2007. Mechanism of pyridine-catalyzed SiO₂ atomic layer deposition studied by fourier transform infrared spectroscopy. *J. Phys. Chem. C* 111, 219–226. <https://doi.org/10.1021/jp0638484>.
- Fukao, K., Tanaka, K., Kiyama, R., et al., 2020. Hydrogels toughened by biominerals providing energy-dissipative sacrificial bonds. *J. Mater. Chem. B* 8, 5184–5188. <https://doi.org/10.1039/D0TB00833H>.
- Guo, L., Han, X., Lei, Y., et al., 2022. Research on nonlinear rheological properties of waxy crude oil based on large amplitude oscillatory shearing. *J. Pet. Sci. Eng.* 213, 110444. <https://doi.org/10.1016/j.petrol.2022.110444>.
- Hackman, I., Hollaway, L., 2006. Epoxy-layered silicate nanocomposites in civil engineering. *Compos. Part A-Appl. Sci.* 37, 1161–1170. <https://doi.org/10.1016/j.compositesa.2005.05.027>.
- Haruna, M.A., Pervaiz, S., Hu, Z., et al., 2019. Improved rheology and high-temperature stability of hydrolyzed polyacrylamide using graphene oxide nanosheet. *J. Appl. Polym. Sci.* 136, 47582. <https://doi.org/10.1002/app.47582>.
- Huang, G., Zhang, H., Zhang, H., et al., 2018. Influence of graphene oxide on the rheological properties of poly(vinyl alcohol)/borate hydrogel. *Chin. J. Appl. Chem.* 35, 767–775. <https://doi.org/10.11944/j.issn.1000-0518.2018.07.170273>.
- Ji, J., Zeng, C., Ke, Y., et al., 2017. Preparation of poly(acrylamide-co-acrylic acid)/silica nanocomposite microspheres and their performance as a plugging material for deep profile control. *J. Appl. Polym. Sci.* 134, 45502. <https://doi.org/10.1002/app.45502>.
- Joshi, D., Maurya, N.K., Kumar, N., et al., 2022. Experimental investigation of silica nanoparticle assisted surfactant and polymer systems for enhanced oil recovery. *J. Pet. Sci. Eng.* 216, 110791. <https://doi.org/10.1016/j.petrol.2022.110791>.
- Karchoubi, F., Afshar Ghotli, R., Pahlevani, H., et al., 2024. New insights into nanocomposite hydrogels; a review on recent advances in characteristics and applications. *Adv. Ind. Eng. Polym. Res.* 7, 54–78. <https://doi.org/10.1016/j.aiepr.2023.06.002>.
- Kumar, D., Lashari, N., Ganat, T., et al., 2022. A review on application of nanoparticles in eOR: performance, mechanisms, and influencing parameters. *J. Mol. Liq.* 353, 118821. <https://doi.org/10.1016/j.molliq.2022.118821>.
- Lele, P., Syed, A.H., Riordon, J., et al., 2018. Deformation of microdroplets in crude oil for rapid screening of enhanced oil recovery additives. *J. Pet. Sci. Eng.* 165, 298–304. <https://doi.org/10.1016/j.petrol.2018.02.009>.
- Li, J., Niu, L., Wu, W., et al., 2020. The reservoir adaptability and oil displacement mechanism of polymer microspheres. *Polymers* 12, 885. <https://doi.org/10.3390/polym12040885>.
- Li, X., Yan, M., Xiao, J., et al., 2023. Ultrafast fabrication of deep eutectic solvent flexible ionic gel with high-transmittance, freeze-resistant and conductivity by frontal

- polymerization. *J. Colloid Interface Sci.* 650, 1382–1392. <https://doi.org/10.1016/j.jcis.2023.07.038>.
- Liu, Z., Wei, P., Qi, Y., et al., 2023. High stretchable and self-healing nanocellulose-poly (acrylic acid) composite hydrogels for sustainable CO₂ shutoff. *Carbohydr. Polym.* 311, 120759. <https://doi.org/10.1016/j.carbpol.2023.120759>.
- Long, W., Zhang, X., Feng, G., et al., 2022. Investigation on chloride binding capacity and stability of Friedel's salt in graphene oxide reinforced cement paste. *Cement Concr. Compos.* 132, 104603. <https://doi.org/10.1016/j.cemconcomp.2022.104603>.
- Lu, C., Yu, C., Yeh, Y., 2021. Engineering nanocomposite hydrogels using dynamic bonds. *Acta Biomater.* 130, 66–79. <https://doi.org/10.1016/j.actbio.2021.05.055>.
- Lu, S., Liu, Q., Li, P., et al., 2024. Preparation and enhancement mechanisms of a novel modified nanographite hybrid polymer gel for profile control in deep reservoirs. *Colloids Surf., A* 681, 132774. <https://doi.org/10.1016/j.colsurfa.2023.132774>.
- Martín, C., Martín-Pacheco, A., Naranjo, A., et al., 2019. Graphene hybrid materials? The role of graphene materials in the final structure of hydrogels. *Nanoscale* 11, 4822–4830. <https://doi.org/10.1039/C8NR09728C>.
- Mekoue Nguela, J., Pereira, Y.F., Sieczkowski, N., et al., 2024. Application of Turbiscan lab® to study the mechanism of fining in red wine using a specific yeast protein extract: a comparison with gelatine. *Colloids Surf., A* 699, 134649. <https://doi.org/10.1016/j.colsurfa.2024.134649>.
- Mohammed, A., Sanjayana, J.G., Duan, W.H., et al., 2015. Incorporating graphene oxide in cement composites: a study of transport properties. *Constr. Build. Mater.* 84, 341–347. <https://doi.org/10.1016/j.conbuildmat.2015.01.083>.
- Narimani, A., Kordnejad, F., Kaur, P., et al., 2023. Synthesis and preparation of poly (AM-co-AMPS)/GO nanocomposites hydrogel as a rheology modifier and fluid loss controller for use in oil well cementing. *J. Dispersion Sci. Technol.* 44, 1738–1749. <https://doi.org/10.1080/01932691.2022.2039687>.
- Natalya, S.A.C., Kadja, G.T.M., Azhari, N.J., et al., 2022. Two-dimensional (2D) nanomaterials for enhanced oil recovery (EOR): a review. *FlatChem* 34, 100383. <https://doi.org/10.1016/j.flatc.2022.100383>.
- Pal, N., Mandal, A., 2020. Oil recovery mechanisms of pickering nanoemulsions stabilized by surfactant-polymer-nanoparticle assemblies: a versatile surface energies' approach. *Fuel* 276, 118138. <https://doi.org/10.1016/j.fuel.2020.118138>.
- Qi, D., Liu, C., Chen, Z., et al., 2015. In situ emulsion copolymerization of methyl methacrylate and butyl acrylate in the presence of SiO₂ with various surface coupling densities. *Colloid Polym. Sci.* 293, 463–471. <https://doi.org/10.1007/s00396-014-3433-3>.
- Qin, D., Gao, W., Zhang, S., et al., 2023. In situ composite of graphene oxide in polyacrylamide to enhance strength of hydrogel with polyethyleneimine as crosslinker. *Geoenergy Sci. Eng.* 231, 212304. <https://doi.org/10.1016/j.geoen.2023.212304>.
- Ren, Y., Zheng, J., Xu, Z., et al., 2018. Application of Turbiscan LAB to study the influence of lignite on the static stability of PCLWS. *Fuel* 214, 446–456. <https://doi.org/10.1016/j.fuel.2017.08.026>.
- Salunkhe, B., Schuman, T., Al Ibrahim, A., et al., 2021. Ultra-high temperature resistant preformed particle gels for enhanced oil recovery. *Chem. Eng. J.* 426, 130712. <https://doi.org/10.1016/j.cej.2021.130712>.
- Seright, R., Brattekas, B., 2021. Water shutoff and conformance improvement: an introduction. *Pet. Sci.* 18, 450–478. <https://doi.org/10.1007/s12182-021-00546-1>.
- Shen, H., Yang, Z., Li, X., et al., 2021. CO₂-responsive agent for restraining gas channeling during CO₂ flooding in low permeability reservoirs. *Fuel* 292, 120306. <https://doi.org/10.1016/j.fuel.2021.120306>.
- Shen, H., Yang, Z., Xiong, Y., et al., 2022. An organic-based amphiphilic Janus polymer nanosheet: synthesis, properties, and microscopic dispersion interpretations. *J. Mol. Liq.* 363, 119822. <https://doi.org/10.1016/j.molliq.2022.119822>.
- Shen, M., Zhang, C., Yan, X., et al., 2024. Research progress and prospects of utilizing carbon-based nanomaterials in enhanced oil recovery. *Adv. Geo-Energy Res.* 14, 201–214. <https://doi.org/10.46690/ager.2024.12.05>.
- Sneh, O., George, S.M., 1995. Thermal stability of hydroxyl groups on a well-defined silica surface. *J. Phys. Chem.* 99, 4639–4647. <https://doi.org/10.1021/j100013a039>.
- Sydansk, R.D., Argabright, P.A., 1988. Conformance Improvement in a Subterranean hydrocarbon-bearing Formation Using a Crosslinked Polymer. Patent. English.
- Tang, W., Zou, C., Liang, H., et al., 2022. The comparison of interface properties on crude oil-water and rheological behavior of four polymeric nanofluids (nano-SiO₂, nano-CaO, GO and CNT) in carbonates for enhanced oil recovery. *J. Pet. Sci. Eng.* 214, 110458. <https://doi.org/10.1016/j.petrol.2022.110458>.
- Tang, X., Kang, W., Zhou, B., et al., 2020. Characteristics of composite microspheres for in-depth profile control in oilfields and the effects of polymerizable silica nanoparticles. *Powder Technol.* 359, 205–215. <https://doi.org/10.1016/j.powtec.2019.09.070>.
- Tang, X., Li, Y., Cao, J., et al., 2023. Adaptability and enhanced oil recovery performance of surfactant–polymer flooding in inverted seven-spot well pattern. *Phys. Fluids* 35. <https://doi.org/10.1063/5.0147806>.
- Tang, X., Li, Y., Du, X., et al., 2025. Internal causes and characteristics of high-pour-point oil non-darcy percolation in porous media under cold damage: an analytical and experimental study. *Fuel* 379, 133001. <https://doi.org/10.1016/j.fuel.2024.133001>.
- Tang, X., Li, Y., Hu, Z., et al., 2024. Rheological behavior of high-pour-point oil: insights into viscosity–temperature prediction and viscoelastic-yielding transition. *Phys. Fluids* 36. <https://doi.org/10.1063/5.0201385>.
- Tisserand, C., Fleury, M., Brunel, L., et al., 2012. Passive microrheology for measurement of the concentrated dispersions stability. In: Starov, V., Griffiths, P. (Eds.), *Progress in Colloid and Polymer Science*. Springer Berlin Heidelberg, Berlin, Heidelberg, pp. 101–105.
- Wang, J., Sun, J., Huang, X., et al., 2024. A salt-responsive amphoteric viscosifier for high-density solid-free completion fluids with high temperature resistance, strong solubility, and high viscosity enhancement. *Geoenergy Sci. Eng.* 243, 213303. <https://doi.org/10.1016/j.geoen.2024.213303>.
- Wang, M., Yang, S., Li, J., et al., 2021. Cold water-flooding in a heterogeneous high-pour-point oil reservoir using computerized tomography scanning: characteristics of flow channel and trapped oil distribution. *J. Pet. Sci. Eng.* 202, 108594. <https://doi.org/10.1016/j.petrol.2021.108594>.
- Wu, C., Zhang, Q., Song, Y., et al., 2017. Microrheology of magnetorheological silicone elastomers during curing process under the presence of magnetic field. *AIP Adv.* 7. <https://doi.org/10.1063/1.5002121>.
- Xiao, L., Hou, J., Sun, J., et al., 2025. Phase behavior of EOR-Oriented dilutable single-phase microemulsions. *Fuel* 381, 133695. <https://doi.org/10.1016/j.fuel.2024.133695>.
- Xiong, R., Guo, J., Kiyiing, W., et al., 2020. Method for judging the stability of asphaltene in crude oil. *ACS Omega* 5, 21420–21427. <https://doi.org/10.1021/acsomega.0c01779>.
- Xu, H., Tan, Y., Rao, W., et al., 2019. Ultra-strong mechanical property and force-driven malleability of water-poor hydrogels. *J. Colloid Interface Sci.* 542, 281–288. <https://doi.org/10.1016/j.jcis.2019.02.015>.
- Xu, M., Cao, Y., Gao, S., 2014. Surface modification of nano-silica with silane coupling agent. *Key Eng. Mater.* 636, 23–27. <https://dx.doi.org/10.4028/www.scientific.net/KEM.636.23>.
- Xu, Z., Zhao, M., Yang, Z., et al., 2024. Novel mussel-inspired high-temperature resistant gel with delayed crosslinking property for ultra-deep reservoir fracturing. *Adv. Funct. Mater.* 34, 2405111. <https://doi.org/10.1002/adfm.202405111>.
- Yang, F., Yang, J., Qiu, S., et al., 2021. Tannic acid enhanced the physical and oxidative stability of chitin particles stabilized oil in water emulsion. *Food Chem.* 346, 128762. <https://doi.org/10.1016/j.foodchem.2020.128762>.
- Yang, W., Shao, B., Liu, T., et al., 2018. Robust and mechanically and electrically self-healing hydrogel for efficient electromagnetic interference shielding. *ACS Appl. Mater. Interfaces* 10, 8245–8257. <https://doi.org/10.1021/acsami.7b18700>.
- Yuan, Y., Zhang, Q., Lin, S., et al., 2025. Water: the soul of hydrogels. *Prog. Mater. Sci.* 148, 101378. <https://doi.org/10.1016/j.pmatsci.2024.101378>.
- Zhang, E., Wang, T., Lian, C., et al., 2013. Robust and thermo-response graphene–PNIPAm hybrid hydrogels reinforced by hectorite clay. *Carbon* 62, 117–126. <https://doi.org/10.1016/j.carbon.2013.06.003>.
- Zhang, H., Li, M., Cai, C., et al., 2024. Minireview on plugging agents for high-conductivity channels in reservoirs. *Energy Fuels* 38, 19139–19155. <https://doi.org/10.1021/acs.energyfuels.4c02545>.
- Zhang, K., Hou, J., Li, Z., 2021. Development and evaluation of an acidic shear-tolerant nanosheet-enhanced cross-linked gel system for hydrofracturing. *Energy Fuels* 35, 7930–7942. <https://doi.org/10.1021/acs.energyfuels.1c00537>.
- Zhang, N., Lu, X., Meng, Y., 2019. Mechanical properties of graphene oxide enhanced polyacrylamide hydrogels. *Polym. Mater. Sci. Eng.* 35, 53–61. <https://doi.org/10.16865/j.cnki.1000-7555.2019.0223>.
- Zhang, Y., Kong, X., Gao, L., et al., 2016. In-situ measurement of viscoelastic properties of fresh cement paste by a microrheology analyzer. *Cement Concr. Res.* 79, 291–300. <https://doi.org/10.1016/j.cemconres.2015.09.020>.
- Zhou, R., Zhang, D., Wei, J., 2022. Experimental investigation on remaining oil distribution and recovery performances after different flooding methods. *Fuel* 322, 124219. <https://doi.org/10.1016/j.fuel.2022.124219>.
- Zhu, D., 2018. Preparation and Gelation Mechanism Study of Polymer Gel Systems for Conformance Improvement in high-temperature Reservoirs. CNKI. <https://link.cnki.net/doi/10.27643/d.cnki.gsybu.2018.000009>.